

OVP Guide to Using Processor Models

Model specific information for RISC-V_RV32IMAC

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Chapter 1

Overview

This document provides the details of an OVP Fast Processor Model variant.

OVP Fast Processor Models are written in C and provide a C API for use in C based platforms. The models also provide a native interface for use in SystemC TLM2 platforms.

The models are written using the OVP VMI API that provides a Virtual Machine Interface that defines the behavior of the processor. The VMI API makes a clear line between model and simulator allowing very good optimization and world class high speed performance. Most models are provided as a binary shared object and also as source. This allows the download and use of the model binary or the use of the source to explore and modify the model.

The models are run through an extensive QA and regression testing process and most model families are validated using technology provided by the processor IP owners. There is a companion document (OVP Guide to Using Processor Models) which explains the general concepts of OVP Fast Processor Models and their use. It is downloadable from the OVPworld website documentation pages.

1.1 Description

RISC-V RV32IMAC 32-bit processor model

1.2 Licensing

This Model is released under the Open Source Apache 2.0

1.3 Extensions

1.3.1 Extensions Enabled by Default

The model has the following architectural extensions enabled, and the corresponding bits in the misa CSR Extensions field will be set upon reset:

misa bit 0: extension A (atomic instructions)

misa bit 2: extension C (compressed instructions)

misa bit 8: RV32I/RV64I/RV128I base integer instruction set

misa bit 12: extension M (integer multiply/divide instructions)

misa bit 18: extension S (Supervisor mode)

misa bit 20: extension U (User mode)

To specify features that can be dynamically enabled or disabled by writes to the misa register in addition to those listed above, use parameter “add_Extensions_mask”. This is a string parameter containing the feature letters to add; for example, value “DV” indicates that double-precision floating point and the Vector Extension can be enabled or disabled by writes to the misa register, if supported on this variant. Parameter “sub_Extensions_mask” can be used to disable dynamic update of features in the same way.

Legacy parameter “misa_Extensions_mask” can also be used. This Uns32-valued parameter specifies all writable bits in the misa Extensions field, replacing any permitted bits defined in the base variant.

Note that any features that are indicated as present in the misa mask but absent in the misa will be ignored. See the next section.

1.3.2 Enabling Other Extensions

The following misa-visible extensions are supported by the model, but not enabled by default in this variant:

misa bit 3: extension D (double-precision floating point)

misa bit 4: RV32E base integer instruction set (embedded)

misa bit 5: extension F (single-precision floating point)

misa bit 7: extension H (hypervisor)

misa bit 13: extension N (user-level interrupts)

misa bit 15: extension P (DSP instructions)

misa bit 21: extension V (vector extension)

misa bit 23: extension X (non-standard extensions present)

To add features from this list to the visible set in the misa register, use parameter “add_Extensions”. This is a string containing identification letters of features to enable; for example, value “DV”

indicates that double-precision floating point and the Vector Extension should be enabled, if they are currently absent and are available on this variant.

Legacy parameter “misa_Extensions” can also be used. This Uns32-valued parameter specifies the reset value for the misa CSR Extensions field, replacing any permitted bits defined in the base variant.

The following implicit extensions are supported by the model, but not enabled by default in this variant:

implicit feature bit 1: extension B (bit manipulation extension)

implicit feature bit 10: extension K (cryptographic)

To add features from this list to the implicitly-enabled set (not visible in the misa register), use parameter “add_implicit_Extensions”. This is a string parameter in the same format as the “add_Extensions” parameter described above.

1.3.3 Disabling Extensions

The following misa-visible extensions are enabled by default in the model and can be disabled:

misa bit 0: extension A (atomic instructions)

misa bit 2: extension C (compressed instructions)

misa bit 12: extension M (integer multiply/divide instructions)

misa bit 18: extension S (Supervisor mode)

misa bit 20: extension U (User mode)

To disable features that are enabled by default, use parameter “sub_Extensions”. This is a string containing identification letters of features to disable; for example, value “DF” indicates that double-precision and single-precision floating point extensions should be disabled, if they are enabled by default on this variant.

1.4 General Features

1.4.1 mtvec CSR

On this variant, the Machine trap-vector base-address register (mtvec) is writable. It can instead be configured as read-only using parameter “mtvec_is_ro”.

Values written to “mtvec” are masked using the value 0xffffffd. A different mask of writable bits may be specified using parameter “mtvec_mask” if required. In addition, on a write that enables vectored interrupt mode, parameter “tvec_align” may be used to specify additional hardware-enforced base address alignment on writes. In this variant, “tvec_align” defaults to 0, implying no additional alignment constraint on writes.

If parameter “mtvec_sext” is True, values written to “mtvec” are sign-extended from the most-significant writable bit. In this variant, “mtvec_sext” is False, indicating that “mtvec” is not

sign-extended.

The initial value of “mtvec” is 0x0. A different value may be specified using parameter “mtvec” if required.

1.4.2 stvec CSR

Values written to “stvec” are masked using the value 0xffffffd. A different mask of writable bits may be specified using parameter “stvec_mask” if required. In addition, on a write that enables vectored interrupt mode, parameter “tvec_align” may be used to specify additional hardware-enforced base address alignment on writes. In this variant, “tvec_align” defaults to 0, implying no additional alignment constraint on writes.

If parameter “stvec_sext” is True, values written to “stvec” are sign-extended from the most-significant writable bit. In this variant, “stvec_sext” is False, indicating that “stvec” is not sign-extended.

1.4.3 Reset

On reset, the model will restart at address 0x0. A different reset address may be specified using parameter “reset_address” or applied using optional input port “reset_addr” if required.

1.4.4 NMI

An NMI input is implemented. To instead specify that NMI is not implemented, set parameter “nmi_absent” to True.

On an NMI, the model will restart at address 0x0; a different NMI address may be specified using parameter “nmi_address” or applied using optional input port “nmi_addr” if required. The cause reported on an NMI is 0x0 by default; a different cause may be specified using parameter “ecode_nmi” or applied using optional input port “nmi_cause” if required.

If parameter “rnmi_version” is not “none”, resumable NMIs are supported, managed by additional CSRs “mnscratch”, “mnepc”, “mncause” and “mnstatus”, following the indicated version of the Resumable NMI extension proposal. In this variant, “rnmi_version” is “none”.

On taking an NMI exception, the mstatus CSR will not be updated. To instead specify that mstatus fields mie and mpie should be updated upon taking an NMI exception, set parameter “nmi_update_mstatus” to True.

On taking an NMI exception, the tcontrol CSR will not be updated. To instead specify that tcontrol fields mte and mpie should be updated upon taking an NMI exception, set parameter “nmi_update_tcontrol” to True.

On taking an NMI exception, the mtval CSR will not be updated. To instead specify that mtval should be written with zero upon taking an NMI exception, set parameter “nmi_zero_mtval” to True.

The NMI input is level-sensitive. To instead specify that the NMI input is latched on the rising

edge of the NMI signal, set parameter “nmi.is.latched” to True.

NMI interrupts are lower priority than Debug and Trigger Module events. To instead specify that NMI interrupts are higher priority, set parameter “nmi.high.priority” to True.

1.4.5 WFI

WFI will halt the processor until an interrupt occurs. It can instead be configured as a NOP by setting parameter “wfi.is.nop” to True.

The nominal time limit for WFI instructions can be set by parameter “TW.time.limit”. In this variant, the time limit is 0 cycles.

Output signal “core.wfi.mode” indicates whether the processor is currently in WFI state.

Input signal “restart.wfi” will cause the processor to restart from WFI state when high.

Parameter “wfi.resume.not.trap” is 0 on this variant, meaning that pending wakeup events when WFI is executed will not prevent a trap occurring. If “wfi.resume.not.trap” is set to 1 then pending wakeup events when WFI is executed will cause the WFI to be treated as a NOP.

1.4.6 cycle CSR

The “cycle” CSR is implemented in this variant. Set parameter “cycle.undefined” to True to instead specify that “cycle” is unimplemented and accesses should cause Illegal Instruction traps.

1.4.7 instret CSR

The “instret” CSR is implemented in this variant. Set parameter “instret.undefined” to True to instead specify that “instret” is unimplemented and accesses should cause Illegal Instruction traps.

1.4.8 hpmcounter CSR

The “hpmcounter” CSRs are implemented in this variant. Set parameter “hpmcounter.undefined” to True to instead specify that “hpmcounter” CSRs are unimplemented and accesses should cause Illegal Instruction traps.

1.4.9 time CSR

The “time” CSR is implemented in this variant. Set parameter “time.undefined” to True to instead specify that “time” is unimplemented and reads of it should cause Illegal Instruction traps. Usually, the value of the “time” CSR should be provided by the platform - see section “Time and Timers” for more information.

1.4.10 mcycle CSR

The “mcycle” CSR is implemented in this variant. Set parameter “mcycle_undefined” to True to instead specify that “mcycle” is unimplemented and accesses should cause Illegal Instruction traps.

1.4.11 minstret CSR

The “minstret” CSR is implemented in this variant. Set parameter “minstret_undefined” to True to instead specify that “minstret” is unimplemented and accesses should cause Illegal Instruction traps.

1.4.12 mhpmcounter CSR

The “mhpmcounter” CSRs are implemented in this variant. Set parameter “mhpmcounter_undefined” to True to instead specify that “mhpmcounter” CSRs are unimplemented and accesses should cause Illegal Instruction traps.

1.4.13 Virtual Memory

This variant supports address translation modes 0 (bare) and 1 (Sv32). Use parameter “Sv_modes” to specify a bit mask of different implemented modes if required; for example, setting “Sv_modes” to $(1 < 0) + (1 < 8)$ indicates that mode 0 (bare) and mode 8 (Sv39) are implemented. These indices correspond to writable values in the satp.MODE CSR field.

A 9-bit ASID is implemented. Use parameter “ASID_bits” to specify a different implemented ASID size if required.

TLB behavior is controlled by parameter “ASIDCacheSize”. If this parameter is 0, then an unlimited number of TLB entries will be maintained concurrently. If this parameter is non-zero, then only TLB entries for up to “ASIDCacheSize” different ASIDs will be maintained concurrently initially; as new ASIDs are used, TLB entries for less-recently used ASIDs are deleted, which improves model performance in some cases. If the model detects that the TLB entry cache is too small (entry ejections are very frequent), it will increase the cache size automatically. In this variant, “ASIDCacheSize” is 8.

Boolean parameter “ignore_non_leaf_DAU” specifies how non-zero D, A and U fields in non-leaf PTE entries are handled. If “ignore_non_leaf_DAU” is False, then using such entries will trigger Page Fault traps. If “ignore_non_leaf_DAU” is True, then such entries will be handled as if D, A and U fields are all zero. In this variant, “ignore_non_leaf_DAU” is 0.

1.4.14 Unaligned Accesses

Unaligned memory accesses are not supported by this variant. Set parameter “unaligned” to “T” to enable such accesses.

Unaligned memory accesses are not supported for AMO instructions by this variant. Set parameter “Zam” to “T” to enable such accesses.

Push/pop instructions will trigger address misaligned exceptions. Set parameter “unalignedPush-Pop” to “T” to enable misaligned accesses just for push/pop instructions.

Address misaligned exceptions are higher priority than page fault or access fault exceptions on this variant. Set parameter “unaligned_low_pri” to “T” to specify that they are lower priority instead.

1.4.15 PMP

16 PMP entries are implemented by this variant. Use parameter “PMP_registers” to specify a different number of PMP entries; set the parameter to 0 to disable the PMP unit.

The PMP grain size (G) is 0, meaning that PMP regions as small as 4 bytes are implemented. Use parameter “PMP_grain” to specify a different grain size if required. (The grain size in bytes is a 64 bit value calculated using the formula $4 \ll \text{PMP_grain}$.)

Unaligned PMP accesses are not decomposed into separate aligned accesses; use parameter “PMP_decompose” to modify this behavior if required.

Parameters to change the write masks for the PMP CSRs are not enabled; use parameter “PMP_maskparams” to modify this behavior if required.

Parameters to change the reset values for the PMP CSRs are not enabled; use parameter “PMP_initialparams” to modify this behavior if required

The total number of PMP registers present is 64, (this includes CSRs for unimplemented PMP regions), as determined by the requirements of the selected priv_version. Set parameter “PMP_csrs” to specify a different total number of PMP registers.

Accesses to unimplemented PMP registers are write-ignored and read as zero on this variant. Set parameter “PMP_undefined” to True to indicate that such accesses should cause Illegal Instruction exceptions instead.

1.4.16 Time and Timers

A RISC-V hart requires an incrementing time source to be available in any of the following cases:

1. The “time” CSR is implemented (“time_undefined” is False);
2. The “Sstc” extension is present (“Sstc” is True);
3. The internal CLINT model is enabled (“CLINT_address” is non-zero).

The time value in this model is implemented by an abstract counter object, which is shared with any other processor models or peripherals that use the same name to define the counter. The name of the counter is specified by the “mtime_counter” parameter (“mtime_counter” by default). The bit width of the counter is specified by the “mtime_bits” parameter (64 by default). The frequency of the counter is specified by the “mtime_Hz” parameter (1e+06Hz by default).

1.5 Compressed Extension

This variant implements the compressed extension with version specified in the References section of this document. Note that parameter “compress_version” can be used to select the required architecture version. See the following sections for detailed information about differences between each supported version.

1.5.1 Compressed Extension Parameters

Parameter “Zca” is used to specify that basic C extension instructions are present. By default, “Zca” is set to 1 in this variant. Updates to this parameter require a commercial product license.

Parameter “Zcf” is used to specify that floating point load/store instructions are present. By default, “Zcf” is set to 1 in this variant. Updates to this parameter require a commercial product license.

Parameter “Zcb” is used to specify that additional simple operation instructions are present. By default, “Zcb” is set to 0 in this variant. Updates to this parameter require a commercial product license.

Parameter “Zcmp” is used to specify that push/pop and double move instructions are present. By default, “Zcmp” is set to 0 in this variant. Updates to this parameter require a commercial product license.

Parameter “Zcmt” is used to specify that table jump instructions are present. By default, “Zcmt” is set to 0 in this variant. Updates to this parameter require a commercial product license.

Parameter “jvt_mask” is used to specify writable bits in the jvt CSR. By default, “jvt_mask” is set to 0xffffffffffc0 in this variant.

1.5.2 Legacy Version 1.10

Legacy encodings with version specified using Zcea, Zceb and Zcee parameters.

1.5.3 Version 0.70.1

All instruction encodings changed from legacy version, with instructions divided into Zca, Zcf, Zcb, Zcmb, Zcmp, Zcmpe and Zcmt subsets.

1.5.4 Version 0.70.5

Version 0.70.5, with these changes compared to version 0.70.1:

- access to jt and jalt instructions is enabled by Smstateen.
- jvt.base is WARL and fewer bits than the maximum can be implemented

1.5.5 Version 1.0.0-RC5.7

Version 1.0.0-RC5.7, with these changes compared to version 0.70.5:

- encodings of jt and jalt instructions changed.
- Zcmb and Zcmpe subsets removed.

1.5.6 Version 1.0

Ratified version 1.0, identical to version 1.0.0-RC5.7.

1.6 Privileged Architecture

This variant implements the Privileged Architecture with version specified in the References section of this document. Note that parameter “priv_version” can be used to select the required architecture version; see the following sections for detailed information about differences between each supported version.

1.6.1 Legacy Version 1.10

1.10 version of May 7 2017.

1.6.2 Version 20190608

Stable 1.11 version of June 8 2019, with these changes compared to version 1.10:

- mcountinhibit CSR defined;
- pages are never executable in Supervisor mode if page table entry U bit is 1;
- mstatus.TW is writable if any lower-level privilege mode is implemented (previously, it was just if Supervisor mode was implemented);

1.6.3 Version 20211203

1.12 draft version of December 3 2021, with these changes compared to version 20190608:

- mstatush, mseccfg, mseccfgh, menvcfg, menvcfgh, senvcfg, henvcfg, henvcfgh and mconfigptr CSRs defined;
- xret instructions clear mstatus.MPRV when leaving Machine mode if new mode is less privileged than M-mode;
- maximum number of PMP registers increased to 64;
- data endian is now configurable.

1.6.4 Version 1.12

Official 1.12 version, identical to 20211203.

1.6.5 Version 1.13

1.13 ratified version of October 17, 2024, with these changes compared to version 20190608:

- misa.MXL is now defined as read-only. The MXL_writable parameter no longer exists when `priv_version >= 1.13`;
- Defined the RV32-only medeleg and hedeleg CSRs;
- Defined hardware-error (19) and software-check (18) exception codes;

1.6.6 Version master

Unstable master version, currently identical to 1.13.

1.7 Unprivileged Architecture

This variant implements the Unprivileged Architecture with version specified in the References section of this document. Note that parameter “user_version” can be used to select the required architecture version; see the following sections for detailed information about differences between each supported version.

1.7.1 Legacy Version 2.2

2.2 version of May 7 2017.

1.7.2 Version 20191213

Stable 20191213-Base-Ratified version of December 13 2019, with these changes compared to version 2.2:

- floating point fmin/fmax instruction behavior modified to comply with IEEE 754-201x.
- numerous other optional behaviors can be separately enabled using Z-prefixed parameters.

1.8 Other Extensions

Other extensions that can be configured are described in this section.

1.8.1 Zmmul

Parameter “Zmmul” is 0 on this variant, meaning that all multiply and divide instructions are implemented. if “Zmmul” is set to 1 then multiply instructions are implemented but divide and remainder instructions are not implemented.

1.8.2 Zicsr

Parameter “Zicsr” is 1 on this variant, meaning that standard CSRs and CSR access instructions are implemented. if “Zicsr” is set to 0 then standard CSRs and CSR access instructions are not implemented and an alternative scheme must be provided as a processor extension.

1.8.3 Zifencei

Parameter “Zifencei” is 1 on this variant, meaning that the fence.i instruction is implemented (but treated as a NOP by the model). if “Zifencei” is set to 0 then the fence.i instruction is not implemented.

1.8.4 Zhintpause

Parameter “Zhintpause” is 1 on this variant, meaning that the pause instruction is implemented. When executed and the PAUSE_time_limit parameter is not 0 it will pause the hart in WFI mode for up to the number of cycles specified by the time limit. When the time limit is 0 PAUSE is a NOP. if “Zhintpause” is set to 0 then the pause instruction is not implemented - instead the opcode will be interpreted as a FENCE instruction.

1.8.5 Zicbom

Parameter “Zicbom” is 0 on this variant, meaning that code block management instructions are undefined. if “Zicbom” is set to 1 then code block management instructions cbo.clean, cbo.flush and cbo.inval are defined.

If Zicbom is present, the cache block size is given by parameter “cmomp_bytes”. The instructions may cause traps if used illegally but otherwise are NOPs in this model.

1.8.6 Zicbop

Parameter “Zicbop” is 0 on this variant, meaning that prefetch instructions are undefined. if “Zicbop” is set to 1 then prefetch instructions prefetch.i, prefetch.r and prefetch.w are defined (but behave as NOPs in this model).

1.8.7 Zicboz

Parameter “Zicboz” is 0 on this variant, meaning that the cbo.zero instruction is undefined. if “Zicboz” is set to 1 then the cbo.zero instruction is defined.

If Zicboz is present, the cache block size is given by parameter “cmoz_bytes”.

1.8.8 Svadu

Parameter “Svadu” is 0 on this variant, meaning that xenvcfg control of hardware update of PTE A and D bits is not implemented. if “Svadu” is set to 1 then xenvcfg control of hardware update of PTE A and D bits is implemented.

1.8.9 Ssdbltrp

Parameter “Ssdbltrp” is 0 on this variant, meaning that S-mode double trap extensions are not implemented. if “Ssdbltrp” is set to 1 then S-mode double trap extensions are implemented.

1.8.10 Smdbltrp

Parameter “Smdbltrp” is 0 on this variant, meaning that M-mode double trap extensions are not implemented. if “Smdbltrp” is set to 1 then M-mode double trap extensions are implemented.

1.8.11 Smstateen

Parameter “Smstateen” is 0 on this variant, meaning that machine/supervisor/hypervisor state enable CSRs are undefined. if “Smstateen” is set to 1 then machine/supervisor/hypervisor state enable CSRs are defined.

Set Ssstateen to True and Smstateen to False to implement only the supervisor/hypervisor state enable CSRs.

1.8.12 Sscofpmf

Parameter “Sscofpmf” is 0 on this variant, meaning that count overflow and mode-based filtering extension CSRs are undefined. if “Sscofpmf” is set to 1 then count overflow and mode-based filtering extension CSRs are defined.

Note that this model implements CSR state only for this extension; hardware performance counters themselves are implementation-specific and not implemented.

1.8.13 Smcntrpmf

Parameter “Smcntrpmf” is 0 on this variant, meaning that cycle and instret mode-based filtering extension CSRs are undefined. if “Smcntrpmf” is set to 1 then cycle and instret mode-based filtering extension CSRs are defined.

extension CSRs are defined.

1.8.14 Smcsrind

Parameter “Smcsrind” is 0 on this variant, meaning that indirect CSR access is not implemented. if “Smcsrind” is set to 1 then indirect CSR access is implemented. If Supervisor mode is implemented, this also implicitly enables Sscsrind.

1.8.15 Smcdeleg

Parameter “Smcdeleg” is 0 on this variant, meaning that supervisor counter delegation is not implemented. if “Smcdeleg” is set to 1 then supervisor counter delegation is implemented. This also implicitly enables Sscfpg.

1.8.16 Ssqosid

Parameter “Ssqosid” is 0 on this variant, meaning that CSR srmcpg is undefined. if “Ssqosid” is set to 1 then CSR srmcpg is defined.

1.8.17 Zawrs

Parameter “Zawrs” is 0 on this variant, meaning that wait-for-reservation-set instructions are not implemented. if “Zawrs” is set to 1 then wait-for-reservation-set instructions are implemented, in which case parameter “TW_time_limit” is used to specify the nominal cycle delay for wrs.nto, and parameter “STO_time_limit” is used to specify the nominal cycle delay for wrs.sto.

1.8.18 Zimop

Parameter “Zimop” is 0 on this variant, meaning that maybe operations are not implemented. if “Zimop” is set to 1 then maybe operations are implemented.

1.8.19 Zcmop

Parameter “Zcmop” is 0 on this variant, meaning that compressed maybe operations are not implemented. if “Zcmop” is set to 1 then compressed maybe operations are implemented.

1.8.20 Zilsd/Zclsd Extensions for Load/Store pair on RV32

Parameter “Zilsd” is 0 on this variant, meaning that 32-bit load/store pair instructions for RV32 are not implemented. if “Zilsd” is set to 1 then 32-bit load/store pair instructions for RV32 are implemented.

Parameter “Zclsd” is 0 on this variant, meaning that 16-bit load/store pair instructions for RV32 are not implemented. if “Zclsd” is set to 1 then 16-bit load/store pair instructions for RV32 are implemented.

1.8.21 Zibi

Parameter “Zibi” is 0 on this variant, meaning that branch-with-immediate instructions are not implemented. if “Zibi” is set to 1 then branch-with-immediate instructions are implemented.

1.9 CLIC

The model can be configured to implement a Core Local Interrupt Controller (CLIC) using parameter “CLICLEVELS”; when non-zero, the CLIC is present with the specified number of interrupt levels (2-256), as described in the RISC-V Core-Local Interrupt Controller specification, and further parameters are made available to configure other aspects of the CLIC. “CLICLEVELS” is zero in this variant, indicating that a CLIC is not implemented.

The model can be configured either to use an internal CLIC model (if parameter “externalCLIC” is False) or to present a net interface to allow the CLIC to be implemented externally in a platform component (if parameter “externalCLIC” is True). When the CLIC is implemented internally, net ports for standard interrupts and additional local interrupts are available. When the CLIC is implemented externally, a net port interface allowing the highest-priority pending interrupt to be delivered is instead present. This is described below.

1.9.1 CLIC Common Parameters

This section describes parameters applicable whether the CLIC is implemented internally or externally. These are:

“CLIC_version”: this defines the version of the CLIC specification that is implemented. See the References section of this document for more information.

“CLICANDBASIC”: this enum parameter indicates whether only the CLICinterrupt controller is present (if “NoBasic”) or both the CLIC and basic (AKA “CLINT”) interrupt controllers are present, with common interrupts shared between the two (if “BasicCommon”) or separate interrupts (if “BasicSeparate”). The default value in this variant is NoBasic.

“local_int_num”: This Uns32 parameter specifies the number of local interrupts implemented in the basic interrupt mode. These interrupts are numbered starting at 16, as the first 16 interrupts in the basic interrupt controller have architecturally defined meanings. The basic local interrupts correspond to bits 16 and above in the mie/mip CSRS. The maximum value for this parameter is 16 for RV32 and 48 for RV64. The default value in this variant is 0.

“NUM_INTERRUPT”: This Uns32 specifies the number of interrupts in the CLIC. If the clicinfo CSR is present then the num_interrupt field will be set to this value. The maximum value for this parameter is 4096. The minimum value for this parameter is 2, although setting it to 0 will set it to the local_int_num value, if that is greater than 2, for backwards compatibility. The default value

in this variant is 0. Note that the `clic_imp_list` can also change the effective value used, by listing interrupt numbers above the specified value.

“CLICXNXTI”: this Boolean parameter indicates whether `xnxti` CSRs are implemented (if True) or unimplemented (if False). The default value in this variant is False.

“CLICXCSW”: this Boolean parameter indicates whether `xscratchsw` and `xscratchswl` CSRs registers are implemented (if True) or unimplemented (if False). The default value in this variant is False.

“`tv_t_undefined`”: this Boolean parameter indicates whether `xtvt` CSRs registers are implemented (if True) or unimplemented (if False). If the registers are unimplemented then the model will use basic mode vectored interrupt semantics based on the `xtvec` CSRs instead of Selective Hardware Vectoring semantics described in the specification. The default value in this variant is False.

“`intthresh_undefined`”: this Boolean parameter indicates whether `xintthresh` CSRs are implemented (if True) or unimplemented (if False). The default value in this variant is False.

“INTTHRESHBITS”: if `xintthresh` CSRs are implemented, this parameter indicates the number of implemented bits in the “th” field. The default value in this variant is 0.

“`mclicbase_undefined`”: this Boolean parameter indicates whether the `mclicbase` CSR register is implemented (if True) or unimplemented (if False); for CLIC versions after 0.9-draft-20220315, this feature is not configurable and this parameter is always True. The default value in this variant is True.

1.9.2 CLIC Internal-Implementation Parameters

This section describes parameters applicable only when the CLIC is implemented internally. These are:

“`mclicbase`”: this parameter specifies the CLIC base address for M-mode interrupts in physical memory. The default value in this variant is 0x0.

“`sclicbase`”: this parameter specifies the CLIC base address for S-mode interrupts in physical memory. The default value in this variant is 0x0.

“CLICCFGMBITS”: this Uns32 parameter indicates the maximum number of bits that can be specified as implemented in `cliccfg.nmbits`, and also indirectly defines `CLICPRIVMODES`. For cores which implement only Machine mode, or which implement only Machine and User modes but not the N extension, the parameter is absent (“CLICCFGMBITS” must be zero in these cases). The default value in this variant is 0.

“CLICCFGLBITS”: this Uns32 parameter indicates the maximum number of bits that can be specified as implemented in `cliccfg.nlbits`. The default value in this variant is 0. See also parameter “`nlbits_valid`” which allows fine-grain specification of legal values in `cliccfg.nlbits`.

“`nlbits_valid`”: this Uns32 parameter is a bitmask of legal values that can be specified as implemented in the WARL `cliccfg.nlbits` field. As an example, a value of 8 in `cliccfg.nlbits` is legal only if bit 8 is set in `nlbits_valid`. The default value of “`nlbits_valid`” in this variant is 0x0.

“CLICSELHVEC”: this Boolean parameter indicates whether Selective Hardware Vectoring is supported (if True) or unsupported (if False). The default value in this variant is False.

“forceCLICmmio”: this Boolean parameter indicates whether the MMIO interface is present when CLIC_version is set to version 0.10-draft-20250317 or later (it is ignored for earlier versions.) As of CLIC version 0.10 the indirect CSR method is required, but the MMIO may optionally be implemented also, by setting this parameter to True. The default value in this variant is False.

“force_clicinfo”: this Boolean parameter forces the clicinfo register to be present. This has no effect when the CLIC_version selection is earlier than 0.9.20221108, where clicinfo is always present. The default value in this variant is False.

1.9.3 CLIC Individual Interrupt Configuration Parameters

The following behaviors are individually configurable for CLIC interrupts using the set of parameters described in this section:

- whether the interrupt is implemented or not. Registers for unimplemented interrupts are read-only zeros, writes ignored, and no input port is created for them.
- whether the clicintattr.MODE field is read-write or read-only, and whether it is initialized to Machine (3) or Supervisor (1).
- whether the clicintattr.TRIG field is read-write or read-only, and whether it is initialized to POSLEVEL (0), POSEDGE (1), NEGLEVEL (2) or NEGEDGE (3).

The following two “clic_X_default” parameters are used to define the default behavior of the clicintattr.MODE and TRIG fields. This will be the behavior for any interrupt not included in any of the “clic_X.list” parameters described later.

“clic_mode_default”: this enum parameter specifies the default behavior for the clicintattr.MODE field for all implemented CLIC interrupts. The supported values are “RW_Machine”, “RW_Supervisor”, “RO_Machine” or “RO_Supervisor”. The default value in this variant is RW_Machine.

“clic_trig_default”: this enum parameter specifies the default behavior for the clicintattr.TRIG field for all implemented CLIC interrupts. The supported values are “RW_” or “RO_”, followed by “POSELEVEL”, “POSEDGE”, “NEGLEVEl” or “NEGEDGE”. The default value in this variant is RW_POSELEVEL.

The following parameters are all comma-separated lists of interrupt numbers (they may be decimal numbers or hex numbers preceded by “0x”). White space is permitted. The largest number allowed is 4095, the highest interrupt number supported by the CLIC architecture. It is a fatal error to specify an invalid list.

The behavior is undefined when the same interrupt number is specified in multiple lists that configure conflicting behavior.

“clic_imp_list” and “clic_unimp_list”: interrupts 0..NUM_INTERRUPTS-1 are implemented by default, and these list parameters are used to refine that default configuration further. Interrupt numbers listed in “clic_imp_list” will be implemented, while those in “clic_unimp_list” will not be implemented. The default values in this variant are clic_imp_list=“” and clic_unimp_list=“”.

“clic_mode_RO_list”, “clic_mode_RW_list”, “clic_mode_M_list” and “clic_mode_S_list”: These are used to override the default behavior of the clicintattr.MODE field, both whether the field is writable

and the initial value for the field. The default values in this variant are `clic_mode_RO_list=""`, `clic_mode_RW_list=""`, `clic_mode_M_list=""`, and `clic_mode_S_list=""`.

`"clic_trig_RO_list"`, `"clic_trig_RW_list"`, `"clic_trig_posedge_list"` and `"clic_trig_negedge_list"`, `"clic_trig_poslevel_list"` and `"clic_trig_neglevel_list"`: These are used to override the default behavior of the `clicintattr.TRIG` field, both whether the field is writable and the initial value for the field. The default values in this variant are `clic_trig_RO_list=""`, `clic_trig_RW_list=""`, `clic_trig_posedge_list=""`, `clic_trig_negedge_list=""`, `clic_trig_poslevel_list=""`, and `clic_trig_neglevel_list=""`.

1.9.4 CLIC Deprecated Parameters

The following parameters are deprecated in favor of the more general-purpose `"clic_X_list"` parameters described above. Everything that is configurable by the deprecated parameters can and should instead be done using those updated parameters.

`"posedge_0.63"` (deprecated): this Uns64 parameter specifies a mask for CLIC interrupts 0 to 63 indicating whether an interrupt is fixed positive-edge triggered. If the corresponding mask bit for interrupt N is 1 then the `clicintattr.trig` field for that interrupt is read-only and initialized appropriately. If zero, then the trig field is either fixed level triggered (see below) or is writable. The default value in this variant is `posedge_0.63=0x0`.

`"posedge_other"` (deprecated): this Boolean parameter indicates whether CLIC interrupts 64 and above are fixed positive-edge triggered. If True then the `clicintattr.trig` fields for these interrupts are read-only and initialized appropriately. If False, then the trig fields are either fixed level triggered (see below) or are writable. The default value in this variant is `posedge_other=False`.

`"poslevel_0.63"` (deprecated): this Uns64 parameter specifies a mask for CLIC interrupts 0 to 63 indicating whether an interrupt is fixed positive-level triggered. If the corresponding mask bit for interrupt N is 1 then the `clicintattr.trig` field for that interrupt is read-only and initialized appropriately. If zero, then the trig field is writable. This is ignored when the interrupt is configured as fixed positive-edge triggered (see above). The default value in this variant is `poslevel_0.63=0x0`.

`"poslevel_other"` (deprecated): this Boolean parameter indicates whether CLIC interrupts 64 and above are fixed positive level triggered. If True then the `clicintattr.trig` fields for these interrupts are read-only and initialized appropriately. If False, then the trig fields are writable. This is ignored when the other interrupts are configured as fixed positive-edge triggered (see above). The default value in this variant is `poslevel_other=False`.

`"CSIP_present"` (deprecated): this Boolean parameter indicates whether the CSIP interrupt is implemented (if True) or unimplemented (if False). If implemented, the interrupt is positive-edge triggered, and has id 12 or 16, depending on the CLIC version selected. The default value in this variant is False.

1.9.5 CLIC External-Implementation Net Port Interface

When the CLIC is externally implemented, net ports are present allowing the external CLIC model to supply the highest-priority pending interrupt and to be notified when interrupts are handled. These are:

“irq_id_i”: this input should be written with the id of the highest-priority pending interrupt.

“irq_lev_i”: this input should be written with the highest-priority interrupt level.

“irq_sec_i”: this 2-bit input should be written with the highest-priority interrupt security state (00:User, 01:Supervisor, 11:Machine).

“irq_shv_i”: this input port should be written to indicate whether the highest-priority interrupt should be direct (0) or vectored (1). If the “tvt_undefined parameter” is False, vectored interrupts will use selective hardware vectoring, as described in the CLIC specification. If “tvt_undefined” is True, vectored interrupts will behave like basic mode vectored interrupts.

“irq_i”: this input should be written with 1 to indicate that the external CLIC is presenting an interrupt, or 0 if no interrupt is being presented.

“irq_ack_o”: this output is written by the model on entry to the interrupt handler (i.e. when the interrupt is taken). It will be written as an instantaneous pulse (i.e. written to 1, then immediately 0).

“irq_id_o”: this output is written by the model with the id of the interrupt currently being handled. It is valid during the instantaneous irq_ack_o pulse.

“sec_lvl_o”: this output signal indicates the current secure status of the processor, as a 2-bit value (00=User, 01:Supervisor, 11=Machine).

1.9.6 CLIC Versions

The CLIC specification has been under active development. To enable simulation of hardware that may be based on an older version of the specification, the model implements behavior for a number of previous versions of the specification. The differing features of these are listed below, in chronological order.

1.9.7 Version 20180831

Legacy version of August 31 2018, compatible with early SiFive cores.

1.9.8 Version 0.9-draft-20191208

Stable 0.9 version of December 8 2019.

1.9.9 Version 0.9-draft-20220315

Stable 0.9 version of March 15 2022, with these changes compared to version 0.9-draft-20191208:

- level-sensitive interrupts may no longer be set pending by software;
- if there is an access exception on table load when Selective Hardware Vectoring is enabled, both xtval and xepc hold the faulting address;

- if the xinhv bit is set, the target address for an xret instruction is obtained by a load from address xepc.

1.9.10 Version 0.9-draft-20221108

Stable 0.9 version of November 8 2022, with these changes compared to version 0.9-draft-20191208:

- vector table reads now require execute permission (not read permission);
- CSIP now has id 16 and is treated as the first local interrupt (previously, it had id 12 and was not considered to be a local interrupt);
- algorithm used for xscratchcsw accesses changed;
- new parameter INTTHRESHBITS allows implemented bits in xintthresh CSRs to be restricted.
- whether interrupt handling is in CLIC or CLINT mode is now determined by mtvec.MODE at all privilege levels (when both modes are supported).
- xintstatus registers have been relocated at read-only CSR addresses (e.g. 0xF46 for mintstatus).
- cliccfg.nvbits field has been removed.
- clicinfo memory-mapped register has been removed.

1.9.11 Version 0.9-draft-20230801

Stable 0.9 version of August 1 2023, with these changes compared to version 0.9-draft-20221108:

- xintstatus registers have been relocated at different read-only CSR addresses (e.g. 0xFB1 for mintstatus);
- 32-bit xcliccfg registers are present on a configuration page preceding interrupt status pages, containing separate unlbts, snlbts and mnlbts fields for each mode;
- if the hart is currently running at privilege mode x, an mret, sret, dret or mnret instruction that changes privilege mode sets xintthresh=0;
- vector table reads now require execute permission (not read permission);
- xepc is forced to table-entry alignment on xRET when inhv is active;
- inhv is now held in exception mode, not interrupt mode.

1.9.12 Version 0.10-draft-20250317

Draft version of March 17 2025, with these changes compared to version 0.9-draft-20230801:

- Indirect CSRs replace the CLIC Memory-Mapped Registers. (new parameter “forceCLICmmio” added to implement MMIO registers in addition to indirect CSRs.)
- CSRs and parameters associated with suclic extension are no longer present. The proposed User mode interrupt N extension is no longer allowed to be enabled when a CLIC version 0.10 or later

is present.

- Add mandatory reset to 0 of mcause.mpipil.
- CLIC is allocated bit 53 in the stateen0 register to block access to the sscllc CSRs.

1.9.13 Version master

Unstable master version as of 17 March 2025 (commit 62092fc)

1.10 Advanced Interrupt Architecture

The model can be configured to implement the Advanced Interrupt Architecture (AIA) interface using Boolean parameter “Smaia”; when True, the AIA interface is present as described in the RISC-V Advanced Interrupt Architecture specification, and further parameters are made available to configure other aspects of the interface. “Smaia” is False in this variant, indicating that the AIA interface is not implemented.

1.11 WorldGuard Extension

The model can be configured to implement the WorldGuard ISA extension using parameter “WG_version”; when not “none”, the WorldGuard ISA extension is present as described in the RISC-V WorldGuard specification, and further parameters are made available to configure other aspects of the extension. “WG_version” is “none” in this variant, indicating that WorldGuard ISA extension is not implemented.

1.12 Zalrsc extension (Load-Reserved/Store-Conditional Locking)

By default, LR/SC locking is implemented automatically by the model and simulator, with a reservation granule defined by the “lr_sc_grain” parameter. Setting “lr_sc_grain” to 1 means the reservation granule is the access size of the lr instruction making the reservation, otherwise the reservation granule is always equal to the “lr_sc_grain” in bytes. Note that any value less than 4 for RV32 or 8 for RV64 will be mapped to 1, and any value not a power of 2 will be mapped to the next lowest power of 2.

In this variant, “lr_sc_grain” is 1, meaning the reservation granule is the access size of the lr instruction making the reservation.

Parameter “amo_aborts_lr_sc” is used to specify whether AMO operations abort any active LR/SC pair. In this variant, “amo_aborts_lr_sc” is False.

Parameter “trap_preserves_lr” is used to specify whether a trap preserves active LR/SC state. In this variant, “trap_preserves_lr” is False.

Parameter “xret_preserves_lr” is used to specify whether an xret instruction preserves active LR/SC state. In this variant, “xret_preserves_lr” is False.

Parameter “lr_sc_use_pa” is used to specify whether LR/SC uses physical addresses, so reservations are supported across virtual aliases. In this variant, “lr_sc_use_pa” is False.

Parameter “lr_sc_abort_self” is used to specify whether LR/SC can be aborted by a store on the same hart that executed the LR instruction. In this variant, “lr_sc_abort_self” is False.

It is possible to implement locking externally to the model in a platform component, using the “LR_address”, “SC_address” and “SC_valid” net ports, as described below.

The “LR_address” output net port is written by the model with the address used by a load-reserved instruction as it executes. This port should be connected as an input to the external lock management component, which should record the address, and also that an LR/SC transaction is active.

The “SC_address” output net port is written by the model with the address used by a store-conditional instruction as it executes. This should be connected as an input to the external lock management component, which should compare the address with the previously-recorded load-reserved address, and determine from this (and other implementation-specific constraints) whether the store should succeed. It should then immediately write the Boolean success/fail code to the “SC_valid” input net port of the model. Finally, it should update state to indicate that an LR/SC transaction is no longer active.

It is also possible to write zero to the “SC_valid” input net port at any time outside the context of a store-conditional instruction, which will mark any active LR/SC transaction as invalid.

Parameter “Zacas” is used to specify whether atomic compare-and-swap instructions are implemented. In this variant, “Zacas” is False.

1.13 Active Atomic Operation Indication

The “AMO_active” output net port is written by the model with a code indicating any current atomic memory operation while the instruction is active. The written codes are:

0: no atomic instruction active

1: AMOMIN active

2: AMOMAX active

3: AMOMINU active

4: AMOMAXU active

5: AMOADD active

6: AMOXOR active

7: AMOOR active

8: AMOAND active

- 9: AMOSWAP active
- 10: AMOCAS.W active (Zacas extension)
- 11: AMOCAS.D active (Zacas extension)
- 12: AMOCAS.Q active (Zacas extension)
- 13: LR active
- 14: SC active

1.14 Interrupts

The “reset” port is an active-high reset input. The processor is halted when “reset” goes high and resumes execution from the reset address specified using the “reset_address” parameter or “reset_addr” port when the signal goes low. The “mcause” register is cleared to zero.

The “nmi” port is an active-high NMI input. The processor resumes execution from the address specified using the “nmi_address” parameter or “nmi_addr” port when the NMI signal goes high. The “mcause” register is cleared to zero.

All other interrupt ports are active high. For each implemented privileged execution level, there are by default input ports for software interrupt, timer interrupt and external interrupt; for example, for Machine mode, these are called “MSWInterrupt”, “MTimerInterrupt” and “MExternalInterrupt”, respectively. When the N extension is implemented, ports are also present for User mode. Parameter “unimp_int_mask” allows the default behavior to be changed to exclude certain interrupt ports. The parameter value is a mask in the same format as the “mip” CSR; any interrupt corresponding to a non-zero bit in this mask will be removed from the processor and read as zero in “mip”, “mie” and “mideleg” CSRs (and Supervisor and User mode equivalents if implemented).

Parameter “posedge_sswi” is used to specify that the SSWInterrupt is edge triggered. This must be set to True if this input is driven by an ACLINT, due to the unique behavior of its SETSSIP register. In this variant, “posedge_sswi” is False.

Parameter “external_int_id” can be used to enable extra interrupt ID input ports on each hart. If the parameter is True then when an external interrupt is taken the value on the ID port is sampled and used to fill the Exception Code field in the relevant “xcause” CSR. For Machine External interrupts, the extra interrupt ID port is called “MExternalInterruptID”; for Supervisor External interrupts, the extra interrupt ID port is called “SEExternalInterruptID”.

The “deferint” port is an active-high artifact input that, when written to 1, prevents any pending-and-enabled interrupt being taken (normally, such an interrupt would be taken on the next instruction after it becomes pending-and-enabled). The purpose of this signal is to enable alignment with hardware models in step-and-compare usage.

1.15 Triggering Exceptions Externally

The following artifact input net port(s) are provided to support triggering exceptions externally (e.g. by a DV test harness or an external bus controller model.)

illegalinstruction - This port is used to immediately trigger an Illegal instruction exception (exception code 2). The exception is triggered when a leading edge (0->1 transition) is seen on the net. If the processor is in a WFI state with a bounded time limit enabled, then the illegal instruction exception follows the conventions for an event causing the hart to resume from WFI mode, namely the EPC will be set to the instruction following the WFI instruction when taking the exception.

hardwareerror - This port is used to immediately trigger a Hardware error exception (exception code 19). The exception is triggered when a leading edge (0->1 transition) is seen on the net. The tval for the exception may be set by driving the artifactdata port (see below) with the desired value immediately before the trigger.

artifactdata - This port is used to provide additional data to be used when another artifact port triggers. When a port that uses artifact data triggers, the most recent value written to the port is sampled, and the value on the port is reset to 0 so a subsequent artifact port trigger will not use a stale value. Do not rely on the previous value being used by subsequent triggers unless it is written again (or it is 0.)

1.16 Debug Mode (Sdext)

The model can be configured to implement Debug mode using parameter “debug_mode”. This implements features described in Chapter 4 of the RISC-V External Debug Support specification with version specified by parameter “debug_version” (see References). Some aspects of this mode are not defined in the specification because they are implementation-specific; the model provides infrastructure to allow implementation of a Debug Module using a custom harness. Features added are described below.

Parameter “debug_mode” can be used to specify four different behaviors, as follows:

1. If set to value “vector”, then operations that would cause entry to Debug mode result in the processor jumping to the address specified by the “debug_address” parameter. It will execute at this address, in Debug mode, until a “dret” instruction causes return to non-Debug mode. Any exception generated during this execution will cause a jump to the address specified by the “dexc_address” parameter.
2. If set to value “interrupt”, then operations that would cause entry to Debug mode result in the processor simulation call (e.g. `opProcessorSimulate`) returning, with a stop reason of `OP_SR_INTERRUPT`. In this usage scenario, the Debug Module is implemented in the simulation harness.
3. If set to value “halt”, then operations that would cause entry to Debug mode result in the processor halting. Depending on the simulation environment, this might cause a return from the simulation call with a stop reason of `OP_SR_HALT`, or debug mode might be implemented by another platform component which then restarts the debugged processor again.
4. If set to value “inject”, then operations that would cause entry to Debug mode result in the processor continuing to execute from the current address in Debug mode. The harness should detect that Debug mode has been entered by monitoring the “DM” integration support register, and inject Debug-mode instructions one at a time using function `opProcessorSimulateInstruction`. Debug mode is exited by either an explicit write of False to the “DM” register or by execution of an injected “dret” instruction, as described by “Debug State Exit” below.

Note that parameter “debug_mode” can be also be set to “none” to specify that Sdext is not implemented.

1.16.1 Debug State Entry

The specification does not define how Debug mode is implemented. In this model, Debug mode is enabled by a Boolean pseudo-register, “DM”. When “DM” is True, the processor is in Debug mode. When “DM” is False, mode is defined by “mstatus” in the usual way.

Entry to Debug mode can be performed in any of these ways:

1. By writing True to register “DM” (e.g. using opProcessorRegWrite) followed by simulation of at least one cycle (e.g. using opProcessorSimulate) - in this case, dcsr.cause will report a cause of trigger (2);
2. By writing a 1 then 0 to net “haltreq” (using opNetWrite) followed by simulation of at least one cycle (e.g. using opProcessorSimulate) - in this case, dcsr.cause will report a cause of haltreq (3);
3. By writing a 1 to net “resethaltreq” (using opNetWrite) while the “reset” signal undergoes a negedge transition, followed by simulation of at least one cycle (e.g. using opProcessorSimulate) - in this case, dcsr.cause will report a cause of resethaltreq (5) or haltreq (3), depending on the value of parameter “no_resethaltreq”;
4. By executing an “ebreak” instruction when Debug mode entry for the current processor mode is enabled by dcsr.ebreakm, dcsr.ebreaks or dcsr.ebreaku - in this case, dcsr.cause will report a cause of ebreak (1);
5. By executing a single instruction when Debug mode entry for the current processor mode is enabled by dcsr.step - in this case, dcsr.cause will report a cause of step (4);
6. By a Trigger Module trigger, when that trigger is configured to enter Debug mode - in this case, dcsr.cause will report a cause of trigger (2).

In all cases, the processor will save required state in “dpc” and “dcsr” and then perform actions described above, depending in the value of the “debug_mode” parameter.

1.16.2 Debug State Exit

Exit from Debug mode can be performed in either of these ways:

1. By writing False to register “DM” (e.g. using opProcessorRegWrite) followed by simulation of at least one cycle (e.g. using opProcessorSimulate);
2. By executing an “dret” instruction when Debug mode.

In both cases, the processor will perform the steps described in section 4.6 (Resume) of the Debug specification.

1.16.3 Debug Registers

When Debug mode is enabled, registers “dcsr” and “dpc” are implemented as described in the specification. Registers “dscratch0” and “dscratch1” may also be implemented (see parameters below). Implemented registers may be manipulated externally by a Debug Module using `opProcessorRegRead` or `opProcessorRegWrite`; for example, the Debug Module could write “dcsr” to enable “ebreak” instruction behavior as described above, or read and write “dpc” to emulate stepping over an “ebreak” instruction prior to resumption from Debug mode.

Parameter “dscratch0_undefined” is used to specify whether the “dscratch0” CSR is undefined, in which case accesses to it trap to Machine mode. In this variant, “dscratch0_undefined” is 0.

Parameter “dscratch1_undefined” is used to specify whether the “dscratch1” CSR is undefined, in which case accesses to it trap to Machine mode. In this variant, “dscratch1_undefined” is 0.

Parameter `dcsr_stepie` is used to specify the behavior of the `dcsr.stepie` field. This is currently configured as read/write, reset to 0.

Parameter `dcsr_stopcount` is used to specify the behavior of the `dcsr.stopcount` field. This is currently configured as read/write, reset to 0.

Parameter `dcsr_stoptime` is used to specify the behavior of the `dcsr.stoptime` field. This is currently configured as read/write, reset to 0.

Parameter `dcsr_mprven` is used to specify the behavior of the `dcsr.mprven` field. This is currently configured as read/write, reset to 0.

1.16.4 Debug Mode Execution

The specification allows execution of code fragments in Debug mode. A Debug Module implementation can cause execution in Debug mode by the following steps:

1. Write the address of a Program Buffer to the program counter using `opProcessorPCSet`;
2. If “debug_mode” is set to “halt”, write 0 to pseudo-register “DMStall” (to leave halted state);
3. If entry to Debug mode was handled by exiting the simulation callback, call `opProcessorSimulate` or `opRootModuleSimulate` to resume simulation.

Debug mode will be re-entered in these cases:

1. By execution of an “ebreak” instruction; or:
2. By execution of an instruction that causes an exception.

In both cases, the processor will either jump to the debug exception address, or return control immediately to the harness, with `stopReason` of `OP_SR_INTERRUPT`, or perform a halt, depending on the value of the “debug_mode” parameter.

1.16.5 Debug Single Step

When in Debug mode, the processor or harness can cause a single instruction to be executed on return from that mode by setting `dcsr.step`. After one non-Debug-mode instruction has been

executed, control will be returned to the harness. The processor will remain in single-step mode until `dcsr.step` is cleared.

1.16.6 Debug Event Priorities

The model supports three different models for determining which debug exception occurs when step, execute address, `resethaltreq` and `haltreq` events are all pending. These options are listed below, with highest-priority event first:

1. when parameter `"debug_priority"="sxh"`: step ->execute address ->`resethaltreq` ->`haltreq`;
2. when parameter `"debug_priority"="shx"`: step ->`resethaltreq` ->`haltreq` ->execute address;
3. when parameter `"debug_priority"="hsx"`: `resethaltreq` ->`haltreq` ->step ->execute address.

1.16.7 Debug Ports

Port `"DM"` is an output signal that indicates whether the processor is in Debug mode

Port `"haltreq"` is a rising-edge-triggered signal that triggers entry to Debug mode (see above).

Port `"resethaltreq"` is a level-sensitive signal that triggers entry to Debug mode after reset (see above).

1.16.8 Debug Mode Versions

Debug mode specification has been under active development. To enable simulation of hardware that may be based on an older version of the specification, the model implements behavior for a number of versions of the specification. The differing features of these are listed below, in chronological order.

1.16.9 Version 0.13.2-DRAFT

0.13.2-DRAFT version of March 22 2019.

1.16.10 Version 0.14.0-DRAFT

0.14.0-DRAFT version of November 6 2020.

1.16.11 Version 1.0.0-STABLE

1.0.0-STABLE version of February 9 2022.

1.16.12 Version 1.0-STABLE

1.0-STABLE version of December 28 2022, with these changes compared to version 1.0.0-STABLE:

- nmi is moved from etrigger to itrigger and is now subject to the mode bits in that trigger.

1.17 Trigger Module (Sdtrig)

This model is configured with a trigger module, implementing a subset of the behavior described in Chapter 5 of the RISC-V External Debug Support specification with version specified by parameter “debug_version” (see References).

1.17.1 Trigger Module Restrictions

The model currently supports tdata1 of type 0, type 2 (mcontrol), type 3 (icount), type 4 (itrigger), type 5 (etrigger) and type 6 (mcontrol6). icount triggers are implemented for a single instruction only, with count hard-wired to 1 and automatic zeroing of mode bits when the trigger fires.

1.17.2 Trigger Module Parameters

Parameter “trigger_num” is used to specify the number of implemented triggers. In this variant, “trigger_num” is 4. Set this parameter to 0 to specify that Sdtrig is not implemented.

Parameter “tinfo” is used to specify the default value of the read-only “tinfo” registers, which indicates the trigger types supported and also version information which controls the behavior of “mcontrol6”. In this variant, “tinfo” is 0x100807c.

In addition, parameters “tinfo0” through “tinfo15” may be used to specify non-default values for individual tinfo registers. See the “Parameters values and limits” section for info on the default setting for each tinfo register.

Parameter “trigger_match” is used to specify the legal “match” values for triggers of types 2 and 6. This parameter is a bitmask with 1 bits corresponding to legal values; for example, a “trigger_match” of 0xd, means that matches of types 0, 2 and 3 are supported. In this variant, “trigger_match” is 0x333f.

Parameter “tdata2_undefined” is used to specify whether the “tdata2” register is undefined, in which case reads of it trap to Machine mode. In this variant, “tdata2_undefined” is 0.

Parameter “tdata3_undefined” is used to specify whether the “tdata3” register is undefined, in which case reads of it trap to Machine mode. In this variant, “tdata3_undefined” is 0.

Parameter “tinfo_undefined” is used to specify whether the “tinfo” register is undefined, in which case reads of it trap to Machine mode. In this variant, “tinfo_undefined” is 0.

Parameter “tcontrol_undefined” is used to specify whether the “tcontrol” register is undefined, in which case accesses to it trap to Machine mode. In this variant, “tcontrol_undefined” is 0.

Parameter “mcontext_undefined” is used to specify whether the “mcontext” register is undefined,

in which case accesses to it trap to Machine mode. In this variant, “mcontext_undefined” is 0.

Parameter “scontext_undefined” is used to specify whether the “scontext” register is undefined, in which case accesses to it trap to Machine mode. In this variant, “scontext_undefined” is 0.

Parameter “mscontext_undefined” is used to specify whether the “mscontext” register is undefined, in which case accesses to it trap to Machine mode. In this variant, “mscontext_undefined” is 0.

Parameter “amo_trigger” is used to specify whether load/store triggers are activated for AMO instructions. In this variant, “amo_trigger” is 0.

Parameter “no_hit” is used to specify whether the “hit” bits in tdata1 are unimplemented. In this variant, “no_hit” is 0.

Parameter “no_sselect_2” is used to specify whether the “sselect” field in “textra32”/“textra64” registers is unable to hold value 2 (indicating match by ASID is not allowed). In this variant, “no_sselect_2” is 0.

Parameter “mcontext_bits” is used to specify the number of writable bits in the “mcontext” register. In this variant, “mcontext_bits” is 6.

Parameter “scontext_bits” is used to specify the number of writable bits in the “scontext” register. In this variant, “scontext_bits” is 16.

Parameter “mvalue_bits” is used to specify the number of writable bits in the “mvalue” field in “textra32”/“textra64” registers; if zero, the “mselect” field is tied to zero. In this variant, “mvalue_bits” is 6.

Parameter “svalue_bits” is used to specify the number of writable bits in the “svalue” field in “textra32”/“textra64” registers; if zero, the “sselect” is tied to zero. In this variant, “svalue_bits” is 16.

Parameter “mcontrol_maskmax” is used to specify the value of field “maskmax” in the “mcontrol” register. In this variant, “mcontrol_maskmax” is 63.

Parameter “mcontrol_addl_comp” is used to specify whether the compare values for trigger types 2 (mcontrol) and 6 (mcontrol6) when select is 0 optionally includes ALL virtual addresses accessed, in addition to the required lowest VA. In this variant, “mcontrol_addl_comp” is False.

1.18 Debug Mask

It is possible to enable model debug messages in various categories. This can be done statically using the “debugflags” parameter, or dynamically using the “debugflags” command. Enabled messages are specified using a bitmask value, as follows:

Value 0x002: enable debugging of PMP and virtual memory state;

Value 0x004: enable debugging of interrupt state;

Value 0x008: enable TLB consistency checking (automatically detect inconsistencies between cached TLB entries and the page table in memory, if these would affect model behavior).

All other bits in the debug bitmask are reserved and must not be set to non-zero values.

1.19 Integration Support

This model implements a number of non-architectural pseudo-registers, commands, and other features to facilitate integration.

1.19.1 Command “setPMA -attributes <attrs>-lo <addr>-hi <addr>”

This command allows PMA attributes to be set for the address range lo:hi. The required attributes are described by the “attrs” string, which can contain any combination of these characters:

“r”: allow read access

“w”: allow write access

“x”: allow execute access

“a”: disallow unaligned accesses

“A”: disallow RVMC_USER1 accesses (often AMO and LR/SC)

“P”: disallow RVMC_USER2 accesses (often push/pop)

“1”: allow 1-byte accesses (if none of 1248 are specified then all are allowed)

“2”: allow 2-byte accesses

“4”: allow 4-byte accesses

“8”: allow 8-byte accesses

<space>, “-”: ignored, use for formatting

The command may be used multiple times, in which case PMA attributes for later commands override those specified for earlier ones where ranges overlap. A common idiom is to deny all access to the entire memory range in the first command before adding back permissions for subregions with subsequent commands.

1.19.2 Command “addLRSCRegion -grainSizeM1 <size-1>-lo <addr>-hi <addr>”

This command allows the LR/SC reservation grain size to be set for the address range lo:hi. For example, a value of 63 for the “grainSizeM1” argument means that the reservation grain size in the indicated address range is 64 bytes. “grainSizeM1” must be one less than a power of two.

The command may be used multiple times, in which case LR/SC reservation grain sizes for later commands override those specified for earlier ones where ranges overlap.

1.19.3 Command “resetLRSCRegion -lo <addr>-hi <addr>”

This command allows the LR/SC reservation grain size to be reset for the address range lo:hi. When reset, the grain size for that region will be the default “lr_sc.grain” specified by the processor configuration.

1.19.4 Command “getCSRIndex -name <name>”

This command returns the index number of a named CSR, or -1 if that CSR does not exist.

1.19.5 Command “listCSRs”

This command lists all implemented CSRs in index order.

1.19.6 CSR Register External Implementation

If parameter “enable_CSR_bus” is True, an artifact 16-bit bus “CSR” is enabled. Slave callbacks installed on this bus can be used to implement modified CSR behavior (use opBusSlaveNew or icmMapExternalMemory, depending on the client API). A CSR with index 0xABC is mapped on the bus at address 0xABC0; as a concrete example, implementing CSR “time” (number 0xC01) externally requires installation of a read callback at address 0xC010 on the CSR bus.

If both read and write callbacks are installed, or if a read callback is installed and the CSR is in the read-only address space, then the read callback will be used to provide the value for both true accesses and for trace and API register read (using opRegRead, etc). However, if only a read callback is installed and the CSR is in the CSR read/write address space then the callback will be used for true register reads **only**; in this case, the **model** CSR implementation will be used for trace and API register read. This idiom allows values to be injected for volatile CSRs without changing fundamental model behavior.

An artifact net, “readcsr”, can also be used to override the value apparently read from a CSR without resorting to the CSR bus. When a CSR is read into a GPR that is not “x0”, this net is written with a value encoding the CSR number (in bits 11:0) and destination GPR number (in bits 20:16). To use this net:

1. Install a net monitor callback on “readcsr” using opNetWriteMonitorAdd;
2. When the callback is activated, extract the encoded CSR and GPR numbers;
3. If the CSR number corresponds to a CSR of interest, find the OP register corresponding to the GPR using opProcessorRegByIndex;
4. Use opProcessorRegWrite to modify the GPR value.

1.19.7 LR/SC Active Address

Artifact register “LRSCAddress” shows the active LR/SC lock address. The register holds all-ones if there is no LR/SC operation active or if LR/SC locking is implemented externally as described above. When parameter “lr_sc_match_size” is True and this is a 64-bit access, the least-significant bit of “LRSCAddress” is one (to indicate a 64-bit access). If parameter “lr_sc_match_size” is False or this is a 32-bit access, the least-significant bit of “LRSCAddress” is zero.

1.19.8 Page Table Walk Introspection

Artifact register “PTWStage” shows the active page table translation stage (0 if no stage active, 1 if HS-stage active, 2 if VS-stage active and 3 if G-stage active). This register is visibly non-zero only in a memory access callback triggered by a page table walk event.

Artifact register “PTWInputAddr” shows the input address of active page table translation. This register is visibly non-zero only in a memory access callback triggered by a page table walk event.

Artifact register “PTWLevel” shows the active level of page table translation (corresponding to index variable “i” in the algorithm described by Virtual Address Translation Process in the RISC-V Privileged Architecture specification). This register is visibly non-zero only in a memory access callback triggered by a page table walk event.

1.19.9 Page Table Walk Query

A banked set of registers provides information about the most recently completed page table walk. There are up to seven banks of registers: bank 0 is for stage 1 walks, bank 1 is for stage 2 walks, and banks 2-6 are for stage 2 walks initiated by stage 1 level 0-4 entry lookups, respectively. Banks 1-6 are present only for processors with the Virtualization Extension. The currently active bank can be set using register “PTWBankSelect”. Register “PTWBankValid” is a bitmask indicating which banks contain valid data: for example, the value 0xb indicates that banks 0, 1 and 3 contain valid data.

Within each bank, there are registers that record addresses and values read during that page table walk. Register “PTWBase” records the table base address, register “PTWInput” contains the input address that starts a walk, register “PTWOutput” contains the result address and register “PTWPgSize” contains the page size (“PTWOutput” and “PTWPgSize” are valid only if the page table walk completes). Registers “PTWAddressL0”-“PTWAddressL4” record the addresses of level 0 to level 4 entries read, respectively. Register “PTWAddressValid” is a bitmask indicating which address registers contain valid data: bits 0-4 indicate “PTWAddressL0”-“PTWAddressL4”, respectively, bit 5 indicates “PTWBase”, bit 6 indicates “PTWInput”, bit 7 indicates both “PTWOutput” and “PTWPgSize”. For example, the value 0xe3 indicates that “PTWBase”, “PTWInput”, “PTWOutput”, “PTWPgSize” and “PTWAddressL0” and “PTWAddressL1” are valid but “PTWAddressL2”-“PTWAddressL4” are not. Registers “PTWValueL0”-“PTWValueL4” contain page table entry values read at level 0 to level 4. Register “PTWValueValid” is a bitmask indicating which value registers contain valid data: bits 0-4 indicate “PTWValueL0”-“PTWValueL4”, respectively.

1.19.10 Artifact Page Table Walks

Registers are also available to enable a simulation environment to initiate an artifact page table walk (for example, to determine the ultimate PA corresponding to a given VA). Register PTWUD initiates a U/HU-mode walk for a load/store address. Register PTWUX initiates a U/HU-mode walk for a fetch address. Registers PTWSD and PTWSX perform corresponding walks for S/HS mode accesses. If Hypervisor mode is implemented, registers PTWVUD and PTWVUX perform corresponding walks for VU mode accesses, and registers PTWVSD and PTWVSX perform corresponding walks for VS mode accesses. Each walk fills the query registers described above.

1.19.11 Page Table Walk Event

Event “pageTableWalk” triggers on completion of any page table walk (including artifact walks). An event callback registered on this event can access the Page Table Walk Query artifact registers described in the previous section.

1.19.12 External Stimulation of Illegal Instruction Trap

Artifact input net port “illegalinstr” allows Illegal Instruction traps to be raised externally. On a rising edge of the signal connected to this port, the hart will immediately take an Illegal Instruction trap with “xepc” set to the current program counter.

As a special case, if the hart is currently stalled by a WFI instruction (“wfi_is_nop” is False), it will be restarted and take either an Illegal Instruction or Virtual Instruction trap, based on the current processor mode and the governing TW bit.

1.19.13 Halt Reason Introspection

An artifact register HaltReason can be read to determine the reason or reasons that a processor is halted. This register is a bitfield, with the following encoding:

0x001: waiting in WFI

0x002: waiting in reset

0x004: waiting for debug

0x008: waiting for reservation set (wrs.*)

0x010: waiting for reservation set (wrs.sto)

0x020: waiting for any pending and enabled interrupt

0x040: waiting for custom reason with WFI semantics

0x080: waiting for custom reason with NMI semantics

0x100: waiting for custom reason (persists over reset)

0x200: waiting because of critical error (Smdbltrp)

An output net of the same name is also present. The net is written whenever the HaltReason changes.

1.20 Instruction Disassembly

This model implements a number of parameters to control instruction disassembly, as shown in trace output.

If parameter “use_hw_reg_names” is True, instruction disassembly shows hardware names x0-x31. If “use_hw_reg_names” is False, ABI names are shown instead.

If parameter “no_pseudo_inst” is True, instruction disassembly always shows true instructions. If “no_pseudo_inst” is False, pseudo-instructions are shown instead where applicable.

If parameter “show_c_prefix” is True, instruction disassembly of 16-bit instructions will include a compressed prefix (e.g. “c.” or “cm.”). If “show_c_prefix” is False, the compressed prefix will be omitted.

1.21 Semihosting

Extension libraries for semihosting of the host computer I/O and related facilities are available. A variety of protocols are supported as described below. These are all in the VLN tree with vendor=riscv.ovpworld.org and library=semihosting.

By default, the pk semihosting (see below) is enabled. To select a different riscv.ovpworld.org semihost library, use the “semihostname” simulator parameter.

To disable semihosting entirely, for example when running an operating system where drivers are implemented along with I/O hardware emulated by platform peripherals, override the “defaultsemihost” processor parameter with the value True.

1.21.1 Proxy Kernel semihosting (default)

The Riscv Proxy Kernel semihosting Execution Environment Interface (EEI) is supported by the riscv.ovpworld.org semihosting library with name=pk. This uses the ecall or ebREAK instruction with register a7 (on RVI) or x5 (on RVE) to hold the call type, while registers a0 thru a6 (on RVI) or x10 thru x15 (on RVE) hold argument values. See <https://github.com/riscv-software-src/riscv-pk> for details.

When using the Riscv GNU gcc toolchain this corresponds to using the command line option `-specs=sim.specs` (which is usually configured as the default behavior.)

1.21.2 Angel-compatible semihosting

An Angel-compatible semihosting Execution Environment Interface (EEI) is supported by the riscv.ovpworld.org semihosting library with name=semihost. This uses the ecall or ebREAK instruction with register a0 to hold the call type, while register a1 contains a pointer to a data block with the argument values for that call type.

When using the Riscv GNU gcc toolchain this corresponds to using the command line option `-specs=semihost.specs`. See libgloss/riscv in <https://sourceware.org/git/?p=newlib-cygwin.git;a=tree> for details.

1.21.3 Newlib interception semihosting

Semihosting that intercepts newlib calls directly is supported by the riscv.ovpworld.org semihosting library with name=riscv32Newlib or riscv64Newlib. Using this semihost library removes any dependency on the EEI protocol implemented by the libgloss layer, but does not provide complete

testing of the toolchain code or provide support for direct environment calls that do not go through newlib functions.

1.22 Limitations

Instruction pipelines are not modeled in any way. All instructions are assumed to complete immediately. This means that instruction barrier instructions (e.g. fence.i) are treated as NOPs, with the exception of any Illegal Instruction behavior, which is modeled.

Caches and write buffers are not modeled in any way. All loads, fetches and stores complete immediately and in order, and are fully synchronous. Data barrier instructions (e.g. fence) are treated as NOPs, with the exception of any Illegal Instruction behavior, which is modeled.

Real-world timing effects are not modeled: all instructions are assumed to complete in a single cycle.

Hardware Performance Monitor registers are not implemented and hardwired to zero.

The TLB is architecturally-accurate but not device accurate. This means that all TLB maintenance and address translation operations are fully implemented but the cache is larger than in the real device.

The internal CLIC does not model the optional interrupt triggers (clicintrig registers.) These registers appear as hard-wired zeros.

1.23 Verification

All instructions have been extensively tested by Imperas, using tests generated specifically for this model and also reference tests from <https://github.com/riscv/riscv-tests>.

Also reference tests have been used from various sources including:

<https://github.com/riscv/riscv-tests>

<https://github.com/ucb-bar/riscv-torture>

The Imperas OVPsim RISC-V models are used in the RISC-V Foundation Compliance Framework as a functional Golden Reference:

<https://github.com/riscv/riscv-compliance>

where the simulated model is used to provide the reference signatures for compliance testing. The Imperas OVPsim RISC-V models are used as reference in both open source and commercial instruction stream test generators for hardware design verification, for example:

<http://valtrix.in/sting> from Valtrix

<https://github.com/google/riscv-dv> from Google

The Imperas OVPsim RISC-V models are also used by commercial and open source RISC-V Core RTL developers as a reference to ensure correct functionality of their IP.

1.24 References

The Model details are based upon the following specifications:

RISC-V Instruction Set Manual, Volume I: User-Level ISA (User Architecture Version 20191213)

RISC-V Instruction Set Manual, Volume II: Privileged Architecture (Privileged Architecture Version 1.13 - ratified 20241017)

RISC-V “C” Compressed Extension (Compressed Architecture Version 1.0)

RISC-V External Debug Support (RISC-V External Debug Support Version 1.0-STABLE as of commit 83483b1 of 21 August 2023 (this is subject to change))

Chapter 2

Configuration

2.1 Location

This model's VLNv is `riscv.ovpworld.org/processor/riscv/1.0`.

The model source is usually at:

`$IMPERAS_HOME/ImperasLib/source/riscv.ovpworld.org/processor/riscv/1.0`

The model binary is usually at:

`$IMPERAS_HOME/lib/$IMPERAS_ARCH/ImperasLib/riscv.ovpworld.org/processor/riscv/1.0`

2.2 GDB Path

The default GDB for this model is: `$IMPERAS_HOME/lib/$IMPERAS_ARCH/gdb/riscv-none-embed-gdb`.

2.3 Semi-Host Library

The default semi-host library file is `riscv.ovpworld.org/semihosting/pk/1.0`

2.4 Processor Endian-ness

This is a LITTLE endian model.

2.5 QuantumLeap Support

This processor is qualified to run in a QuantumLeap enabled simulator.

2.6 Processor ELF code

The ELF code supported by this model is: `0xf3`.

Chapter 3

All Variants in this model

This model has these variants

Variant	Description
RV32I	
RV32IM	
RV32IMC	
RV32IMCZ _{ce}	
RV32IMAC	(described in this document)
RV32G	
RV32GC	
RV32GCZ _{fnx}	
RV32GCB	
RV32GCH	
RV32GCK	
RV32GCN	
RV32GCV	
RV32E	
RV32EC	
RV32EM	
RV64I	
RV64IM	
RV64IMC	
RV64IMCZ _{ce}	
RV64IMAC	
RV64G	
RV64GC	
RV64GCZ _{fnx}	
RV64GCB	
RV64GCH	
RV64GCK	
RV64GCN	
RV64GCV	
RVB32I	
RVB32E	

RVB64I	
RVI20U32mandatory	
RVI20U32	
RVI20U64mandatory	
RVI20U64	
RVA20U64mandatory	
RVA20U64	
RVA20S64mandatory	
RVA20S64	
RVA22U64mandatory	
RVA22U64	
RVA22S64mandatory	
RVA22S64	
RVA23U64mandatory	
RVA23U64	
RVA23S64mandatory	
RVA23S64	
RVB23U64mandatory	
RVB23U64	
RVB23S64mandatory	
RVB23S64	

Table 3.1: All Variants in this model

Chapter 4

Bus Master Ports

This model has these bus master ports.

Name	min	max	Connect?	Description
INSTRUCTION	32	34	mandatory	Instruction bus
DATA	32	34	optional	Data bus

Table 4.1: Bus Master Ports

Chapter 5

Bus Slave Ports

This model has no bus slave ports.

Chapter 6

Net Ports

This model has these net ports.

Name	Type	Connect?	Description
reset	input	optional	Reset
reset_addr	input	optional	Externally-applied reset address
nmi	input	optional	NMI
nmi_cause	input	optional	Externally-applied NMI cause
nmi_addr	input	optional	Externally-applied NMI address
SSWInterrupt	input	optional	Supervisor software interrupt
MSWInterrupt	input	optional	Machine software interrupt
STimerInterrupt	input	optional	Supervisor timer interrupt
MTimerInterrupt	input	optional	Machine timer interrupt
SExternalInterrupt	input	optional	Supervisor external interrupt
MExternalInterrupt	input	optional	Machine external interrupt
irq_ack_o	output	optional	Interrupt acknowledge (pulse)
irq_id_o	output	optional	Acknowledged interrupt id (valid during irq_ack_o pulse)
sec_lvl_o	output	optional	Current privilege level
LR_address	output	optional	Port written with effective address for LR instruction
SC_address	output	optional	Port written with effective address for SC instruction
SC_valid	input	optional	SC_address valid input signal
AMO_active	output	optional	Port written with code indicating active AMO
illegalinstr	input	optional	Artifact signal causing Illegal Instruction exception on rising edge
hardwareerror	input	optional	Artifact signal causing Hardware Error exception on rising edge
artifactdata	input	optional	Artifact data value. Sampled on hardwareerror artifact port triggers
deferint	input	optional	Artifact signal causing interrupts to be held off while high

coverpoint	output	optional	Artifact port written with coverage point identifier
readcsr	output	optional	Artifact port written with CSR/GPR information when CSR is read
HaltReason	output	optional	Artifact halt reason indication
core_wfi_mode	output	optional	WFI active indication
restart_wfi	input	optional	Artifact signal causing restart from WFI state when high

Table 6.1: Net Ports

Chapter 7

FIFO Ports

This model has no FIFO ports.

Chapter 8

Formal Parameters

Name	Type	Description
Fundamental		
user_version	Enumeration	Specify required User Architecture version
	2.2	User Architecture Version 2.2
	2.3	Deprecated and equivalent to 20191213
	20190305	Deprecated and equivalent to 20191213
	20191213	User Architecture Version 20191213
priv_version	Enumeration	Specify required Privileged Architecture version
	1.10	Privileged Architecture Version 1.10
	1.11	Privileged Architecture Version 1.11 - ratified 20190608
	1.12	Privileged Architecture Version 1.12 - ratified 20211203
	1.13	Privileged Architecture Version 1.13 - ratified 20241017
	master	Privileged Architecture Master Branch as of commit 2c07aa2 (this is subject to change)
	20190405	Deprecated and equivalent to 1.11
	20190608	Deprecated and equivalent to 1.11
	20211203	Deprecated and equivalent to 1.12
Smepmp_version	Enumeration	Specify required Smepmp Architecture version
	none	Smepmp not implemented
	0.9.5	Smepmp version 0.9.5 (deprecated and identical to 1.0)
	1.0	Smepmp version 1.0
numHarts	Uns32	Specify the number of hart contexts in a multiprocessor
enable_host_atomics	Boolean	Enable use of host atomic instructions to emulate load/store exclusive operations (not architecturally accurate, accelerates parallel simulation)
enable_expanded	Boolean	Specify that 48-bit and 64-bit expanded instructions are supported
endianFixed	Boolean	Specify that data endianness is fixed (mstatus.{MBE,SBE,UBE} fields are read-only)
misa_MXL	Uns32	Override default value of misa.MXL
misa_Extensions	Uns32	Override default value of misa.Extensions
add_Extensions	String	Add extensions specified by letters to misa.Extensions (for example, specify “VD” to add V and D features)

sub_Extensions	String	Remove extensions specified by letters from misa.Extensions (for example, specify “VD” to remove V and D features)
misa_Extensions_mask	Uns32	Override mask of writable bits in misa.Extensions
add_Extensions_mask	String	Add extensions specified by letters to mask of writable bits in misa.Extensions (for example, specify “VD” to add V and D features)
sub_Extensions_mask	String	Remove extensions specified by letters from mask of writable bits in misa.Extensions (for example, specify “VD” to remove V and D features)
add_implicit_Extensions	String	Add extensions specified by letters to implicitly-present extensions not visible in misa.Extensions
sub_implicit_Extensions	String	Remove extensions specified by letters from implicitly-present extensions not visible in misa.Extensions
Compressed_Extension		
compress_version	Enumeration	Specify required Compressed Architecture version
	legacy	Compressed Architecture absent or legacy version
	0.70.1	Compressed Architecture Version 0.70.1
	0.70.5	Compressed Architecture Version 0.70.5
	1.0.0-RC5.7	Compressed Architecture Version 1.0.0-RC5.7
	1.0	Compressed Architecture Version 1.0
Zca	Boolean	Specify that Zca is implemented
Zcb	Boolean	Specify that Zcb is implemented
Zcf	Boolean	Specify that Zcf is implemented
Zcmp	Boolean	Specify that Zcmp is implemented
Zcmt	Boolean	Specify that Zcmt is implemented
Zcmop	Boolean	Specify that Zcmop is implemented
Debug_Extension		
debug_version	Enumeration	Specify required Debug Architecture version
	0.13.2-DRAFT	RISC-V External Debug Support Version 0.13.2-DRAFT
	0.14.0-DRAFT	RISC-V External Debug Support Version 0.14.0-DRAFT
	1.0.0-STABLE	RISC-V External Debug Support Version 1.0.0-STABLE
	1.0-STABLE	RISC-V External Debug Support Version 1.0-STABLE as of commit 83483b1 of 21 August 2023 (this is subject to change)
debug_mode	Enumeration	Specify how Debug mode is implemented (value ‘none’ implies Sdext is not implemented)
	none	Debug mode not implemented
	vector	Debug mode implemented by execution at vector
	interrupt	Debug mode implemented by interrupt
	halt	Debug mode implemented by halt
	inject	Debug mode implemented using injected instructions
Interrupts_Exceptions		
rnmi_version	Enumeration	Specify required RNMI Architecture version
	none	RNMI not implemented
	0.2.1	RNMI version 0.2.1
	0.4.nmie1	RNMI version 0.4, except mnstatus.nmie=1 at reset
	0.4	RNMI version 0.4
mtvec_is_ro	Boolean	Specify whether mtvec CSR is read-only

tvec.mode_rsvd	Enumeration	Specify behavior of writes of reserved values to xTVEC.mode field (ignored when CLIC.version < 20191208)
	TVEC.M.KEEP	Attempt to set mode to 2 keeps previous value
	TVEC.M.CLIC.VECTOR	Attempt to set mode to 2 will be mapped to 3 (Vectored CLIC mode)
tvec.align	Uns32	Specify additional hardware-enforced alignment on writes to mtvec/stvec/utvec that set a non-direct mode (i.e. bit 0 of the write is 1)
ecode.mask	Uns64	Specify hardware-enforced mask of writable bits in xcause.ExceptionCode
ecode.nmi	Uns64	Specify xcause.ExceptionCode for NMI
nmi.absent	Boolean	Specify whether NMI input is absent
nmi.is_latched	Boolean	Specify whether NMI input is latched on rising edge (if False, it is level-sensitive)
nmi.high_priority	Boolean	Specify whether NMI input is higher priority than Debug and Trigger Module events
nmi.update_mstatus	Boolean	Specify whether an NMI exception updates the mstatus CSR mie and mpie fields
nmi.update_tcontrol	Boolean	Specify whether an NMI exception updates the tcontrol CSR mte and mpie fields
nmi.zero_mtval	Boolean	Specify whether an NMI exception writes 0 to the mtval CSR
mtval.is_ro	Boolean	Specify whether mtval CSR is read-only
tval.zero	Boolean	Specify whether mtval/stval/vstval/utval are hard wired to zero
tval.mask	Uns64	Specify write mask for mtval/stval/vstval/utval. Ignored if tval.zero is True, 0 maps to -1ULL
tval.zero.ecodes.mask	Uns64	Specify mask of exception codes that always set mtval to 0. Not used for interrupts, ignored if tval.zero is True
tval.zero.ebreak	Boolean	Specify whether mtval/stval/vstval/utval are set to zero by an ebreak
tval.ii.code	Boolean	Specify whether mtval/stval contain faulting instruction bits on illegal instruction exception
trap.preserves_lr	Boolean	Whether a trap preserves active LR/SC state
xret.preserves_lr	Boolean	Whether an xret instruction preserves active LR/SC state
reset.address	Uns64	Override reset vector address
nmi.address	Uns64	Override NMI vector address
CLINT.address	Uns64	Specify base address of internal CLINT model (or 0 for no CLINT)
local.int_num	Uns32	Specify number of local interrupts (excludes standard set 0 to 15 if basic interrupt mode is present)
unimp.int_mask	Uns64	Specify mask of unimplemented interrupts (e.g. 1 <= 9 indicates Supervisor external interrupt unimplemented)
force.mideleg	Uns64	Specify mask of interrupts always delegated to lower-priority execution level from Machine execution level
force.sideleg	Uns64	Specify mask of interrupts always delegated to User execution level from Supervisor execution level
no.ideleg	Uns64	Specify mask of interrupts that cannot be delegated to lower-priority execution levels
no.edeleg	Uns64	Specify mask of exceptions that cannot be delegated to lower-priority execution levels

external_int_id	Boolean	Whether to add nets allowing External Interrupt ID codes to be forced
posedge_sswi	Boolean	Specify that Supervisor SW Interrupt input is edge-triggered (required when driven from ACLINT)
Fast Interrupt		
CLIC_version	Enumeration	Specify required CLIC version
	20180831	CLIC Version 20180831 (commit 6d369ab)
	0.9-draft-20191208	CLIC Version 0.9-draft-20191208 (commit f9ab734)
	0.9-draft-20220315	CLIC Version 0.9-draft-20220315 (commit 45a2e91)
	0.9-draft-20221108	CLIC Version 0.9-draft-20221108 (commit fdcd068)
	0.9-draft-20230801	CLIC Version 0.9-draft-20230801 (commit 97a2d57)
	0.10-draft-20250317	CLIC Version 0.10-draft-20250317 (commit 62092fc)
	master	CLIC Master Branch as of commit 62092fc (this is subject to change)
CLICLEVELS	Uns32	Specify number of interrupt levels implemented by CLIC, or 0 if CLIC absent
WorldGuard		
WG_version	Enumeration	Specify required WorldGuard version
	none	WorldGuard not implemented
	0.4	WorldGuard Version 0.4
Memory		
lr_sc_constraint	Enumeration	Specify memory constraint for LR/SC instructions
	none	Memory access not constrained
	user1	Memory access constrained by MEM.CONSTRAINT.USER1
	user2	Memory access constrained by MEM.CONSTRAINT.USER2
	user3	Memory access constrained by MEM.CONSTRAINT.USER3
	user4	Memory access constrained by MEM.CONSTRAINT.USER4
amo_constraint	Enumeration	Specify memory constraint for AMO instructions
	none	Memory access not constrained
	user1	Memory access constrained by MEM.CONSTRAINT.USER1
	user2	Memory access constrained by MEM.CONSTRAINT.USER2
	user3	Memory access constrained by MEM.CONSTRAINT.USER3
	user4	Memory access constrained by MEM.CONSTRAINT.USER4
push_pop_constraint	Enumeration	Specify memory constraint for PUSH/POP instructions
	none	Memory access not constrained
	user1	Memory access constrained by MEM.CONSTRAINT.USER1
	user2	Memory access constrained by MEM.CONSTRAINT.USER2
	user3	Memory access constrained by MEM.CONSTRAINT.USER3
	user4	Memory access constrained by MEM.CONSTRAINT.USER4
updatePTEA	Boolean	Specify whether hardware update of PTE A bit is supported (ignored when parameter Svadu is True)
updatePTED	Boolean	Specify whether hardware update of PTE D bit is supported (ignored when parameter Svadu is True)

unaligned_low_pri	Boolean	Specify whether address misaligned exceptions are lower priority than page or access fault exceptions
unaligned	Boolean	Specify whether the processor supports unaligned memory accesses
Zam	Boolean	Specify whether the processor supports unaligned memory accesses for AMO instructions
unalignedPushPop	Boolean	Specify whether the processor supports unaligned memory accesses for push/pop (deprecated, ignored when unaligned=True)
aligned_uncached_PBMT	Boolean	Specify whether pages with non-zero PBMT require aligned access (Svpbmt extension only)
amo_aborts_lr_sc	Boolean	Specify whether AMO operations abort any active LR/SC pair
ASID_bits	Uns32	Specify the number of implemented ASID bits
lr_sc_grain	Uns32	Specify byte granularity of LR/SC lock region (constrained to a power of two)
lr_sc_use_pa	Boolean	Whether LR/SC uses physical addresses, so reservations are supported across virtual aliases
lr_sc_abort_self	Boolean	Whether LR/SC can be aborted by a store on the same hart that executed the LR instruction
ignore_non_leaf_DAU	Boolean	Whether non-zero D, A and U bits in non-leaf PTEs are ignored (if False, a trap is taken)
MPU_grain	Uns32	Specify Ssmpu region granularity, G (0 =>4 bytes, 1 =>8 bytes, etc)
MPU_registers	Uns32	Specify the number of implemented Ssmpu address registers
MPU_decompose	Boolean	Whether unaligned Ssmpu accesses are decomposed into separate aligned accesses
Sv_modes	Uns32	Specify bit mask of implemented address translation modes (e.g. (1<<0)+(1<<8) indicates “bare” and “Sv39” modes may be selected in satp.MODE)
Simulation Artifact		
leaf_hart_prefix	String	Specify string name prefix for harts in a cluster
mtime_counter	String	Specify name of shared mtime counter
mtime_Hz	Double	Specify mtime counter frequency
mtime_bits	Uns32	Specify mtime counter bit width
use_hw_reg_names	Boolean	Specify whether to use hardware register names x0-x31 and f0-f31 instead of ABI register names
no_pseudo_inst	Boolean	Specify whether pseudo-instructions should not be reported in trace and disassembly
show_c_prefix	Boolean	Specify whether compressed instruction prefix should be reported in trace and disassembly
verbose	Boolean	Specify verbose output messages
traceVolatile	Boolean	Specify whether volatile registers (e.g. minstret) should be shown in change trace
wfi_restart	Boolean	Specify whether to automatically restart from WFI state when model is simulated
enable_CSR_bus	Boolean	Add artifact CSR bus port, allowing CSR registers to be externally implemented
CSR_remap	String	Comma-separated list of CSR number mappings, each of the form <csrName>=<number>
ASID_cache_size	Uns32	Specify the number of different ASIDs for which TLB entries are cached; a value of 0 implies no limit
Instruction_CSR_Behavior		
wfi_is_nop	Boolean	Specify whether WFI should be treated as a NOP (if not, halt while waiting for interrupts)

wfi_resume_not_trap	Boolean	Specify whether pending wakeup events should cause WFI to be treated as a NOP instead of taking a trap
TW_time_limit	Uns32	Specify nominal cycle timeout for instructions controlled by mstatus.TW
PAUSE_time_limit	Uns32	Specify nominal cycle timeout for PAUSE instruction
counteren_mask	Uns32	Specify hardware-enforced mask of writable bits in mcounteren/scounteren registers
scounteren_zero_mask	Uns32	Specify hardware-enforced mask of always-zero bits in scounteren register
noinhibit_mask	Uns32	Specify hardware-enforced mask of always-zero bits in mcountinhibit register
cycle_undefined	Boolean	Specify that the cycle CSR is undefined
mcycle_undefined	Boolean	Specify that the mcycle CSR is undefined
time_undefined	Boolean	Specify that the time CSR is undefined
instret_undefined	Boolean	Specify that the instret CSR is undefined
minstret_undefined	Boolean	Specify that the minstret CSR is undefined
hpmcounter_undefined	Boolean	Specify that the hpmcounter CSRs are undefined
mhpmcounter_undefined	Boolean	Specify that the mhpmcounter CSRs are undefined
CSR Masks		
mtvec_mask	Uns64	Specify hardware-enforced mask of writable bits in mtvec register
stvec_mask	Uns64	Specify hardware-enforced mask of writable bits in stvec register
jvt_mask	Uns64	Specify hardware-enforced mask of writable bits in Zcmt jvt register
tdata1_mask	Uns64	Specify hardware-enforced mask of writable bits in Trigger Module tdata1 register
mip_mask	Uns64	Specify hardware-enforced mask of writable bits in mip register
sip_mask	Uns64	Specify hardware-enforced mask of writable bits in sip register
envcfg_mask	Uns64	Specify hardware-enforced mask of writable bits in envcfg registers
mtvec_sext	Boolean	Specify whether mtvec is sign-extended from most-significant bit
stvec_sext	Boolean	Specify whether stvec is sign-extended from most-significant bit
csrindCustom	Boolean	Force MSB of *iselect CSRs to be writable to support custom extensions using Smcsrind
Trigger		
tdata2_undefined	Boolean	Specify that the tdata2 CSR is undefined
tdata3_undefined	Boolean	Specify that the tdata3 CSR is undefined
tinfo_undefined	Boolean	Specify that the tinfo CSR is undefined
tcontrol_undefined	Boolean	Specify that the tcontrol CSR is undefined
mcontext_undefined	Boolean	Specify that the mcontext CSR is undefined
scontext_undefined	Boolean	Specify that the scontext CSR is undefined
mscontext_undefined	Boolean	Specify that the mscontext CSR is undefined (Debug Version 0.14.0 and later)
amo_trigger	Boolean	Specify whether AMO load/store operations activate triggers
no_hit	Boolean	Specify that tdata1.hit* bits are unimplemented
no_sselect_2	Boolean	Specify that textra.sselect=2 is not supported (no trigger match by ASID)

trigger_num	Uns32	Specify the number of implemented hardware triggers (0 implies Sdtrig is not implemented)
mask_tselect	Boolean	Whether values written to tselect are masked according to the trigger_num setting
tinfo	Uns32	Override tinfo register (for all triggers)
tinfo0	Uns32	Specify value of read-only tinfo for trigger 0
tinfo1	Uns32	Specify value of read-only tinfo for trigger 1
tinfo2	Uns32	Specify value of read-only tinfo for trigger 2
tinfo3	Uns32	Specify value of read-only tinfo for trigger 3
tinfo4	Uns32	Specify value of read-only tinfo for trigger 4
tinfo5	Uns32	Specify value of read-only tinfo for trigger 5
tinfo6	Uns32	Specify value of read-only tinfo for trigger 6
tinfo7	Uns32	Specify value of read-only tinfo for trigger 7
tinfo8	Uns32	Specify value of read-only tinfo for trigger 8
tinfo9	Uns32	Specify value of read-only tinfo for trigger 9
tinfo10	Uns32	Specify value of read-only tinfo for trigger 10
tinfo11	Uns32	Specify value of read-only tinfo for trigger 11
tinfo12	Uns32	Specify value of read-only tinfo for trigger 12
tinfo13	Uns32	Specify value of read-only tinfo for trigger 13
tinfo14	Uns32	Specify value of read-only tinfo for trigger 14
tinfo15	Uns32	Specify value of read-only tinfo for trigger 15
trigger_match	Uns32	Specify legal “match” values for triggers of type 2 and 6 (bitmask)
mcontext_bits	Uns32	Specify the number of implemented bits in mcontext
scontext_bits	Uns32	Specify the number of implemented bits in scontext
mvalue_bits	Uns32	Specify the number of implemented bits in textra.mvalue (if zero, textra.mselect is tied to zero)
svalue_bits	Uns32	Specify the number of implemented bits in textra.svalue (if zero, textra.sselect is tied to zero)
mcontrol_maskmax	Uns32	Specify mcontrol.maskmax value
chain_tval	Enumeration	Specify which trigger provides xtval when triggers are chained
	first	first matching trigger provides xtval
	last	last matching trigger provides xtval
	first_non_epc	first matching trigger provides xtval (prefer non-epc)
	last_non_epc	last matching trigger provides xtval (prefer non-epc)
mcontrol_addl_comp	Boolean	whether mcontrol compare for select=0 includes recommended additional values
PMP Configuration		
PMP_grain	Uns32	Specify PMP region granularity, G (0 =>4 bytes, 1 =>8 bytes, etc)
PMP_registers	Uns32	Specify the number of supported PMP regions
PMP_csrs	Uns32	Specify the number of PMP address CSRs (are RAZ/WI if corresponding region is not implemented)
PMP_max_page	Uns32	Specify the maximum size of PMP region to map if non-zero (may improve performance; constrained to a power of two)
PMP_decompose	Boolean	Whether unaligned PMP accesses are decomposed into separate aligned accesses
PMP_undefined	Boolean	Whether accesses to unimplemented PMP registers are undefined (if True) or write ignored and zero (if False)
PMP_R0W1	Enumeration	Specify behavior of PMPiCFG RWX field on illegal write with R=0 and W=1

	RWX_00X	set R=0 and W=0, modify X
	RWX_00P	set R=0 and W=0, preserve X
	RWX_11X	set R=1 and W=1, modify X
	RWX_PPX	preserve previous R and W, modify X
	RWX_PPP	preserve previous RWX
	RWX_000	set RWX=000
PMP_A_RSVD	Enumeration	Specify behavior of writes of reserved values to PMPiCFG.A
	PMP_A_NONE	No reserved values for PMPiCFG.A
	PMP_NA4_OFF	Attempts to write NA4 mode to PMPiCFG.A will be mapped to OFF
PMP_maskparams	Boolean	Enable parameters to change the read-only masks for PMP CSRs
PMP_initialparams	Boolean	Enable parameters to change the reset values for PMP CSRs
Ssmp	Boolean	Specify that Ssmp is implemented
mpmpdeleg_min	Uns32	Specify minimum value of mpmpdeleg (if Ssmp enabled)
mpmpdeleg_max	Uns32	Specify maximum value of mpmpdeleg (if Ssmp enabled)
Other Extensions		
Svnapot_page_mask	Uns64	Specify mask of implemented Svnapot intermediate page sizes (e.g. 1<<16 means 64KiB contiguous regions are supported)
Smstateen	Boolean	Specify that Smstateen is implemented (ignored when priv_version <1.12)
Ssstateen	Boolean	Specify that Ssstateen is implemented (implied when Smstateen is True, ignored when priv_version <1.12)
Smcsrind	Boolean	Specify that Smcsrind is implemented (plus Sscsrind if S extension is implemented)
Sstc	Boolean	Specify that Sstc is implemented
Sscfpmf	Boolean	Specify that Sscfpmf is implemented
Ssdbltrp	Boolean	Specify that Ssdbltrp is implemented
Smdbltrp	Boolean	Specify that Smdbltrp is implemented
Smcncrpmf	Boolean	Specify that Smcncrpmf is implemented
Smcdeleg	Boolean	Specify that Smcdeleg is implemented
Svpbmt	Boolean	Specify that Svpbmt is implemented
Ssnpm	Enumeration	Specify Ssnpm implemented modes
	none	pointer masking not implemented
	48	PM=XLEN-48 is implemented
	57	PM=XLEN-57 is implemented
	48_57	PM=XLEN-48 and PM=XLEN-57 are implemented
Smnpm	Enumeration	Specify Smnpm implemented modes
	none	pointer masking not implemented
	48	PM=XLEN-48 is implemented
	57	PM=XLEN-57 is implemented
	48_57	PM=XLEN-48 and PM=XLEN-57 are implemented
Smmmpm	Enumeration	Specify Smmmpm implemented modes
	none	pointer masking not implemented
	48	PM=XLEN-48 is implemented
	57	PM=XLEN-57 is implemented
	48_57	PM=XLEN-48 and PM=XLEN-57 are implemented
Svinval	Boolean	Specify that Svinval is implemented
Svadu	Boolean	Specify that Svadu is implemented
Ssqosid	Boolean	Specify that Ssqosid is implemented

Zihintntl	Boolean	Specify that Zihintntl is implemented (instruction decode only, implemented as NOP)
Zihintpause	Boolean	Specify that Zihintpause is implemented
Zicond	Boolean	Specify that Zicond is implemented
Zicsr	Boolean	Specify that Zicsr is implemented
Zifencei	Boolean	Specify that Zifencei is implemented
Zicbom	Boolean	Specify that Zicbom is implemented
Zicbop	Boolean	Specify that Zicbop is implemented
Zicboz	Boolean	Specify that Zicboz is implemented
Zimop	Boolean	Specify that Zimop is implemented
Zicfiss	Boolean	Specify that Zicfiss is implemented
Zicfilp	Boolean	Specify that Zicfilp is implemented
Zilsd	Boolean	Specify that Zilsd is implemented
Zibi	Boolean	Specify that Zibi is implemented
Zclsd	Boolean	Specify that Zclsd is implemented
ZxlsdPair	Boolean	Specify that Zilsd/Zclsd (if implemented) use two 4-byte memory operations
Zaamo	Boolean	Specify that Zaamo is implemented
Zalrsc	Boolean	Specify that Zalrsc is implemented
Zacas	Boolean	Specify that Zacas is implemented
Zabha	Boolean	Specify that Zabha is implemented
Zawrs	Boolean	Specify that Zawrs is implemented
Zmmul	Boolean	Specify that Zmmul is implemented
Bit_Manipulation_Extension		
misa.B.Zba.Zbb.Zbs	Boolean	Specify that misa.B is present if Zba, Zbb and Zbs are all implemented (from bit manipulation version 1.0.0 only; ignored for previous versions)
Zba	Boolean	Specify that Zba is implemented
Zbb	Boolean	Specify that Zbb is implemented
Zbc	Boolean	Specify that Zbc is implemented
Zbe	Boolean	Specify that Zbe is implemented (ignored if version 1.0.0)
Zbf	Boolean	Specify that Zbf is implemented (ignored if version 1.0.0)
Zbm	Boolean	Specify that Zbm is implemented (ignored if version 1.0.0)
Zbp	Boolean	Specify that Zbp is implemented (ignored if version 1.0.0)
Zbr	Boolean	Specify that Zbr is implemented (ignored if version 1.0.0)
Zbs	Boolean	Specify that Zbs is implemented
Zbt	Boolean	Specify that Zbt is implemented (ignored if version 1.0.0)
CSR_Defaults		
mvendorid	Uns64	Override mvendorid register
marchid	Uns64	Override marchid register
mimpid	Uns64	Override mimpid register
mhartid	Uns64	Override mhartid register (or first mhartid of an incrementing sequence if this is an SMP variant)
mconfigptr	Uns64	Override mconfigptr register
mtvec	Uns64	Override mtvec register
mseccfg	Uns64	Override mseccfg register
menvcfg	Uns64	Override menvcfg register
Floating_Point		
mstatus_FS_zero	Boolean	Specify that mstatus.FS is hard-wired to zero
AIA_Interrupts		

Smaia	Boolean	Specify that Smaia CSRs are present
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Table 8.1: Parameters that can be set in: Hart

8.1 Parameters with enumerated types

8.1.1 Parameter user_version

Set to this value	Description
2.2	User Architecture Version 2.2
2.3	Deprecated and equivalent to 20191213
20190305	Deprecated and equivalent to 20191213
20191213	User Architecture Version 20191213

Table 8.2: Values for Parameter user_version

8.1.2 Parameter priv_version

Set to this value	Description
1.10	Privileged Architecture Version 1.10
1.11	Privileged Architecture Version 1.11 - ratified 20190608
1.12	Privileged Architecture Version 1.12 - ratified 20211203
1.13	Privileged Architecture Version 1.13 - ratified 20241017
master	Privileged Architecture Master Branch as of commit 2c07aa2 (this is subject to change)
20190405	Deprecated and equivalent to 1.11
20190608	Deprecated and equivalent to 1.11
20211203	Deprecated and equivalent to 1.12

Table 8.3: Values for Parameter priv_version

8.1.3 Parameter compress_version

Set to this value	Description
legacy	Compressed Architecture absent or legacy version
0.70.1	Compressed Architecture Version 0.70.1
0.70.5	Compressed Architecture Version 0.70.5
1.0.0-RC5.7	Compressed Architecture Version 1.0.0-RC5.7
1.0	Compressed Architecture Version 1.0

Table 8.4: Values for Parameter compress_version

8.1.4 Parameter debug_version

Set to this value	Description
0.13.2-DRAFT	RISC-V External Debug Support Version 0.13.2-DRAFT
0.14.0-DRAFT	RISC-V External Debug Support Version 0.14.0-DRAFT
1.0.0-STABLE	RISC-V External Debug Support Version 1.0.0-STABLE
1.0-STABLE	RISC-V External Debug Support Version 1.0-STABLE as of commit 83483b1 of 21 August 2023 (this is subject to change)

Table 8.5: Values for Parameter debug_version

8.1.5 Parameter rnmi_version

Set to this value	Description
none	RNMI not implemented
0.2.1	RNMI version 0.2.1
0.4_nmie1	RNMI version 0.4, except mnstatus.nmie=1 at reset
0.4	RNMI version 0.4

Table 8.6: Values for Parameter rnmi_version

8.1.6 Parameter Smepmp_version

Set to this value	Description
none	Smepmp not implemented
0.9.5	Smepmp version 0.9.5 (deprecated and identical to 1.0)
1.0	Smepmp version 1.0

Table 8.7: Values for Parameter Smepmp_version

8.1.7 Parameter CLIC_version

Set to this value	Description
20180831	CLIC Version 20180831 (commit 6d369ab)
0.9-draft-20191208	CLIC Version 0.9-draft-20191208 (commit f9ab734)
0.9-draft-20220315	CLIC Version 0.9-draft-20220315 (commit 45a2e91)
0.9-draft-20221108	CLIC Version 0.9-draft-20221108 (commit fdcd068)
0.9-draft-20230801	CLIC Version 0.9-draft-20230801 (commit 97a2d57)
0.10-draft-20250317	CLIC Version 0.10-draft-20250317 (commit 62092fc)
master	CLIC Master Branch as of commit 62092fc (this is subject to change)

Table 8.8: Values for Parameter CLIC_version

8.1.8 Parameter WG_version

Set to this value	Description
none	WorldGuard not implemented
0.4	WorldGuard Version 0.4

Table 8.9: Values for Parameter WG_version

8.1.9 Parameter debug_mode

Set to this value	Description
none	Debug mode not implemented
vector	Debug mode implemented by execution at vector
interrupt	Debug mode implemented by interrupt
halt	Debug mode implemented by halt
inject	Debug mode implemented using injected instructions

Table 8.10: Values for Parameter debug_mode

8.1.10 Parameter lr_sc_constraint

Set to this value	Description
none	Memory access not constrained
user1	Memory access constrained by MEM_CONSTRAINT_USER1
user2	Memory access constrained by MEM_CONSTRAINT_USER2
user3	Memory access constrained by MEM_CONSTRAINT_USER3
user4	Memory access constrained by MEM_CONSTRAINT_USER4

Table 8.11: Values for Parameter lr_sc_constraint

8.1.11 Parameter amo_constraint

Set to this value	Description
none	Memory access not constrained
user1	Memory access constrained by MEM_CONSTRAINT_USER1
user2	Memory access constrained by MEM_CONSTRAINT_USER2
user3	Memory access constrained by MEM_CONSTRAINT_USER3
user4	Memory access constrained by MEM_CONSTRAINT_USER4

Table 8.12: Values for Parameter amo_constraint

8.1.12 Parameter push_pop_constraint

Set to this value	Description
none	Memory access not constrained
user1	Memory access constrained by MEM_CONSTRAINT_USER1
user2	Memory access constrained by MEM_CONSTRAINT_USER2
user3	Memory access constrained by MEM_CONSTRAINT_USER3
user4	Memory access constrained by MEM_CONSTRAINT_USER4

Table 8.13: Values for Parameter push_pop_constraint

8.1.13 Parameter tvec_mode_rsvd

Set to this value	Description
TVEC_M_KEEP	Attempt to set mode to 2 keeps previous value
TVEC_M_CLIC_VECTOR	Attempt to set mode to 2 will be mapped to 3 (Vectored CLIC mode)

Table 8.14: Values for Parameter tvec_mode_rsvd

8.1.14 Parameter chain_tval

Set to this value	Description
first	first matching trigger provides xtval
last	last matching trigger provides xtval
first_non_epc	first matching trigger provides xtval (prefer non-epc)
last_non_epc	last matching trigger provides xtval (prefer non-epc)

Table 8.15: Values for Parameter chain_tval

8.1.15 Parameter PMP_R0W1

Set to this value	Description
RWX_00X	set R=0 and W=0, modify X
RWX_00P	set R=0 and W=0, preserve X
RWX_11X	set R=1 and W=1, modify X
RWX_PPX	preserve previous R and W, modify X
RWX_PPP	preserve previous RWX
RWX_000	set RWX=000

Table 8.16: Values for Parameter PMP_R0W1

8.1.16 Parameter PMP_A_RSVD

Set to this value	Description
PMP_A_NONE	No reserved values for PMPiCFG.A
PMP_NA4.OFF	Attempts to write NA4 mode to PMPiCFG.A will be mapped to OFF

Table 8.17: Values for Parameter PMP_A_RSVD

8.1.17 Parameter Ssnpm

Set to this value	Description
none	pointer masking not implemented
48	PM=XLEN-48 is implemented
57	PM=XLEN-57 is implemented
48_57	PM=XLEN-48 and PM=XLEN-57 are implemented

Table 8.18: Values for Parameter Ssnpm

8.1.18 Parameter Smnpm

Set to this value	Description
none	pointer masking not implemented
48	PM=XLEN-48 is implemented
57	PM=XLEN-57 is implemented
48_57	PM=XLEN-48 and PM=XLEN-57 are implemented

Table 8.19: Values for Parameter Smnpm

8.1.19 Parameter Smmpm

Set to this value	Description
none	pointer masking not implemented
48	PM=XLEN-48 is implemented
57	PM=XLEN-57 is implemented
48_57	PM=XLEN-48 and PM=XLEN-57 are implemented

Table 8.20: Values for Parameter Smmpm

8.2 Parameter values and limits

These are the formal parameter limits and actual parameter values

Name	Min	Max	Default	Actual
Fundamental				
variant				RV32IMAC
user_version			20191213	20191213
priv_version			1.13	1.13
Smepmp_version			none	none
numHarts	0	0x400	0	0
endian			none	none
enable_host_atomics			f	f
enable_expanded			f	f
endianFixed			f	f
misa_MXL	1	2	1	1
misa_Extensions	0	0x3ffffff	0x141105	0x141105
add_Extensions				
sub_Extensions				
misa_Extensions_mask	0	0x3ffffff	0x1105	0x1105
add_Extensions_mask				
sub_Extensions_mask				
add_implicit_Extensions				
sub_implicit_Extensions				
Compressed_Extension				
compress_version			1.0	1.0
Zca			t	t
Zcb			f	f
Zcf			t	t
Zcmp			f	f
Zcmt			f	f
Zcmop			f	f
Debug_Extension				
debug_version			1.0-STABLE	1.0-STABLE
debug_mode			none	none
Interrupts_Exceptions				
rnmi_version			none	none
mtvec_is_ro			f	f
tvec_mode_rsvd			TVEC_M_KEEP	TVEC_M_KEEP
tvec_align	0	0x10000	0	0
ecode_mask	0	0xffffffffffff	0x7fffffff	0x7fffffff
ecode_nmi	0	0xffffffffffff	0	0
nmi_absent			f	f
nmi_is_latched			f	f
nmi_high_priority			f	f
nmi_update_mstatus			f	f

nmi_update_tcontrol			f	f
nmi_zero_mtval			f	f
mtval_is_ro			f	f
tval_zero			f	f
tval_mask	0	0xffffffffffff	0	0
tval_zero_ecodes_mask	0	0xffffffffffff	0	0
tval_zero_ebreak			f	f
tval_ii_code			t	t
trap_preserves_lr			f	f
xret_preserves_lr			f	f
reset_address	0	0xffffffffffff	0	0
nmi_address	0	0xffffffffffff	0	0
CLINT_address	0	0xffffffffffff	0	0
local_int_num	0	16	0	0
unimp_int_mask	0	0xffffffffffff	0	0
force_mideleg	0	0xffffffffffff	0	0
force_sideleg	0	0xffffffffffff	0	0
no_ideleg	0	0xffffffffffff	0	0
no_e deleg	0	0xffffffffffff	0	0
external_int_id			f	f
posedge_sswi			f	f
Fast Interrupt				
CLIC_version			0.10-draft-20250317	0.10-draft-20250317
CLICLEVELS	0	0x100	0	0
WorldGuard				
WG_version			none	none
Memory				
lr_sc_constraint			user1	user1
amo_constraint			user1	user1
push_pop_constraint			user2	user2
updatePTEA			f	f
updatePTED			f	f
unaligned_low_pri			f	f
unaligned			f	f
Zam			f	f
unalignedPushPop			f	f
aligned_uncached_PBMT			f	f
amo_aborts_lr_sc			f	f
ASID_bits	0	9	9	9
lr_sc_grain	1	0x10000	1	1
lr_sc_use_pa			f	f
lr_sc_abort_self			f	f
ignore_non_leaf_DAU			f	f
MPU_grain	0	29	0	0
MPU_registers	0	64	0	0

MPU_decompose			f	f
Sv_modes	0	0xffff	3	3
Simulation Artifact				
leaf_hart_prefix			hart	hart
mtime_counter			mtime_counter	mtime_counter
mtime_Hz	0.000000e+00	1.000000e+09	1.000000e+06	1.000000e+06
mtime_bits	0	64	64	64
use_hw_reg_names			f	f
no_pseudo_inst			f	f
show_c_prefix			f	f
verbose			f	f
traceVolatile			f	f
wfi_restart			f	f
enable_CSR_bus			f	f
CSR_remap				
ASID_cache_size	0	0x100	8	8
Instruction_CSR_Behavior				
wfi_is_nop			f	f
wfi_resume_not_trap			f	f
TW_time_limit	0	0xffffffff	0	0
PAUSE_time_limit	0	0xffffffff	0	0
counteren_mask	0	0xffffffff	0xffffffff	0xffffffff
scounteren_zero_mask	0	0xffffffff	0	0
noinhibit_mask	0	0xffffffff	0	0
cycle_undefined			f	f
mcycle_undefined			f	f
time_undefined			f	f
instret_undefined			f	f
minstret_undefined			f	f
hpmcounter_undefined			f	f
mhpmcounter_undefined			f	f
CSR Masks				
mtvec_mask	0	0xffffffffffffff	0	0
stvec_mask	0	0xffffffffffffff	0	0
jvt_mask	0	0xffffffffffffff	0xffffffffffffc0	0xffffffffffffc0
tdata1_mask	0	0xffffffffffffff	0xffffffffffffff	0xffffffffffffff
mip_mask	0	0xffffffffffffff	0x337	0x337
sip_mask	0	0xffffffffffffff	0x103	0x103
envcfg_mask	0	0xffffffffffffff	0	0
mtvec_sext			f	f
stvec_sext			f	f
csrindCustom			f	f
Trigger				
tdata2_undefined			f	f
tdata3_undefined			f	f

tinfo_undefined			f	f
tcontrol_undefined			f	f
mcontext_undefined			f	f
scontext_undefined			f	f
mscontext_undefined			f	f
amo_trigger			f	f
no_hit			f	f
no_sselect_2			f	f
trigger_num	0	255	4	4
mask_tselect			f	f
tinfo	0	0x100ffff	0x100807c	0x100807c
tinfo0	0	0xffffffff	0x100807c	0x100807c
tinfo1	0	0xffffffff	0x100807c	0x100807c
tinfo2	0	0xffffffff	0x100807c	0x100807c
tinfo3	0	0xffffffff	0x100807c	0x100807c
tinfo4	0	0xffffffff	0x100807c	0x100807c
tinfo5	0	0xffffffff	0x100807c	0x100807c
tinfo6	0	0xffffffff	0x100807c	0x100807c
tinfo7	0	0xffffffff	0x100807c	0x100807c
tinfo8	0	0xffffffff	0x100807c	0x100807c
tinfo9	0	0xffffffff	0x100807c	0x100807c
tinfo10	0	0xffffffff	0x100807c	0x100807c
tinfo11	0	0xffffffff	0x100807c	0x100807c
tinfo12	0	0xffffffff	0x100807c	0x100807c
tinfo13	0	0xffffffff	0x100807c	0x100807c
tinfo14	0	0xffffffff	0x100807c	0x100807c
tinfo15	0	0xffffffff	0x100807c	0x100807c
trigger_match	1	0xffff	0x333f	0x333f
mcontext_bits	0	32	6	6
scontext_bits	0	32	16	16
mvalue_bits	0	6	6	6
svalue_bits	0	16	16	16
mcontrol_maskmax	0	63	63	63
chain_tval			first	first
mcontrol_addl_comp			f	f
PMP Configuration				
PMP_grain	0	29	0	0
PMP_registers	0	64	16	16
PMP_csrs	0	64	0	0
PMP_max_page	0	0xffffffff	0	0
PMP_decompose			f	f
PMP_undefined			f	f
PMP_R0W1			RWX_00X	RWX_00X
PMP_A_RSVD			PMP_A_NONE	PMP_A_NONE
PMP_maskparams			f	f

PMP_initialparams			f	f
Sspmp			f	f
mpmpdeleg_min	0	127	0	0
mpmpdeleg_max	0	127	0	0
Other Extensions				
Svnapot_page_mask	0	0xffffffffffff	0	0
Smstateen			f	f
Ssstateen			f	f
Smcsrind			f	f
Sstc			f	f
Sscofpmf			f	f
Ssdbltrp			f	f
Smdbltrp			f	f
Smentrpmf			f	f
Smcdeleg			f	f
Svpbmt			f	f
Ssnpm			none	none
Smnpm			none	none
Smmpm			none	none
Svinval			f	f
Svadu			f	f
Ssqosid			f	f
Zihintntl			f	f
Zihintpause			t	t
Zicond			f	f
Zicsr			t	t
Zifencei			t	t
Zicbom			f	f
Zicbop			f	f
Zicboz			f	f
Zimop			f	f
Zicfiss			f	f
Zicfilp			f	f
Zilsd			f	f
Zibi			f	f
Zclsd			f	f
ZxlsdPair			f	f
Zaamo			t	t
Zalrsc			t	t
Zacas			f	f
Zabha			f	f
Zawrs			f	f
Zmmul			f	f
Bit Manipulation Extension				
misa.B.Zba.Zbb.Zbs			f	f

Zba			t	t
Zbb			t	t
Zbc			t	t
Zbe			t	t
Zbf			t	t
Zbm			t	t
Zbp			t	t
Zbr			t	t
Zbs			t	t
Zbt			t	t
CSR Defaults				
mvendorid	0	0xffffffffffff	0	0
marchid	0	0xffffffffffff	0	0
mimpid	0	0xffffffffffff	0	0
mhartid	0	0xffffffffffff	0	0
mconfigptr	0	0xffffffffffff	0	0
mtvec	0	0xffffffffffff	0	0
mseccfg	0	0xffffffffffff	0	0
menvcfg	0	0xffffffffffff	0	0
Floating Point				
mstatus_FS_zero			f	f
AIA Interrupts				
Smaia			f	f

Table 8.21: Parameter values and limits

Chapter 9

Execution Modes

Mode	Code	Description
User	0	User mode
Supervisor	1	Supervisor mode
Machine	3	Machine mode

Table 9.1: Modes implemented in: Hart

Chapter 10

Exceptions

Exception	Code	Description
InstructionAddressMisaligned	0	Fetch from unaligned address
InstructionAccessFault	1	No access permission for fetch
IllegalInstruction	2	Undecoded, unimplemented or disabled instruction
Breakpoint	3	EBREAK instruction executed
LoadAddressMisaligned	4	Load from unaligned address
LoadAccessFault	5	No access permission for load
StoreAMOAddressMisaligned	6	Store/atomic memory operation at unaligned address
StoreAMOAccessFault	7	No access permission for store/atomic memory operation
EnvironmentCallFromUMode	8	ECALL instruction executed in User mode
EnvironmentCallFromSMode	9	ECALL instruction executed in Supervisor mode
EnvironmentCallFromMMode	11	ECALL instruction executed in Machine mode
InstructionPageFault	12	Page fault at fetch address
LoadPageFault	13	Page fault at load address
StoreAMOPageFault	15	Page fault at store/atomic memory operation address
SoftwareCheck	18	Software check
HardwareError	19	Hardware Error
SSWInterrupt	65	Supervisor software interrupt
MSWInterrupt	67	Machine software interrupt
STimerInterrupt	69	Supervisor timer interrupt
MTimerInterrupt	71	Machine timer interrupt
SExternalInterrupt	73	Supervisor external interrupt
MExternalInterrupt	75	Machine external interrupt
GenericNMI	4294967295	Generic NMI

Table 10.1: Exceptions implemented in: Hart

Chapter 11

Hierarchy of the model

A CPU core may be configured to instance many processors of a Symmetrical Multi Processor (SMP). A CPU core may also have sub elements within a processor, for example hardware threading blocks.

OVP processor models can be written to include SMP blocks and to have many levels of hierarchy. Some OVP CPU models may have a fixed hierarchy, and some may be configured by settings in a configuration register. Please see the register definitions of this model.

This model documentation shows the settings and hierarchy of the default settings for this model variant.

11.1 Level 1: Hart

This level in the model hierarchy has 9 commands.

This level in the model hierarchy has 5 register groups:

Group name	Registers
Core	33
User_Control_and_Status	64
Supervisor_Control_and_Status	13
Machine_Control_and_Status	200
Integration_support	66

Table 11.1: Register groups

This level in the model hierarchy has no children.

Chapter 12

Model Commands

A Processor model can implement one or more **Model Commands** available to be invoked from the simulator command line, from the OP API or from the Imperas Multiprocessor Debugger.

12.1 Level 1: Hart

12.1.1 addLRSCRegion

Add LR/SC region with the given grain size

Argument	Type	Description
-grainSizeM1	Uns64	grain size, minus 1 (e.g. specify 63 for 64-bit reservation grain)
-hi	Uns64	high address
-lo	Uns64	low address

Table 12.1: addLRSCRegion command arguments

12.1.2 debugflags

show or modify the processor debug flags

Argument	Type	Description
-get	Boolean	print current processor flags value
-mask	Boolean	print valid debug flag bits
-set	Int32	new processor flags (only flags 0x00000006 can be modified)

Table 12.2: debugflags command arguments

12.1.3 dumpTLB

12.1.3.1 Argument description

Show TLB contents

12.1.4 getCSRIndex

Return index for a named CSR (or -1 if no matching CSR)

Argument	Type	Description
-name	String	CSR name

Table 12.3: getCSRIndex command arguments

12.1.5 isync

specify instruction address range for synchronous execution

Argument	Type	Description
-addresshi	Uns64	end address of synchronous execution range
-addresslo	Uns64	start address of synchronous execution range

Table 12.4: isync command arguments

12.1.6 itrace

enable or disable instruction tracing

Argument	Type	Description
-access	String	show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-after	Uns64	apply after this many instructions
-enable	Boolean	enable instruction tracing
-full	Boolean	turn on all trace features
-instructioncount	Boolean	include the instruction number in each trace
-memory	String	(Alias for access). show memory accesses by this instruction. Argument can be any combination of X (execute), A (load or store access) and S (system)
-mode	Boolean	show processor mode changes
-off	Boolean	disable instruction tracing
-on	Boolean	enable instruction tracing
-processorname	Boolean	Include processor name in all trace lines
-registerchange	Boolean	show registers changed by this instruction
-registers	Boolean	show registers after each trace

Table 12.5: itrace command arguments

12.1.7 listCSRs

12.1.7.1 Argument description

List all CSRs in index order

12.1.8 resetLRSCRegion

Reset LR/SC region (use configured lr_sc_grain in this range)

Argument	Type	Description
-hi	Uns64	high address
-lo	Uns64	low address

Table 12.6: resetLRSCRegion command arguments

12.1.9 setPMA

Set PMA region permissions and legal access sizes

Argument	Type	Description
-attributes	String	region attributes (string containing r, w, x, a, A, P, 1, 2, 4 or 8)
-hi	Uns64	high address
-lo	Uns64	low address

Table 12.7: setPMA command arguments

Chapter 13

Registers

13.1 Level 1: Hart

13.1.1 Core

Registers at level:1, type:Hart group:Core

Name	Bits	Initial-Hex	RW	Description
zero	32	0	r-	
ra	32	0	rw	
sp	32	0	rw	stack pointer
gp	32	0	rw	
tp	32	0	rw	
t0	32	0	rw	
t1	32	0	rw	
t2	32	0	rw	
s0	32	0	rw	
s1	32	0	rw	
a0	32	0	rw	
a1	32	0	rw	
a2	32	0	rw	
a3	32	0	rw	
a4	32	0	rw	
a5	32	0	rw	
a6	32	0	rw	
a7	32	0	rw	
s2	32	0	rw	
s3	32	0	rw	
s4	32	0	rw	
s5	32	0	rw	
s6	32	0	rw	
s7	32	0	rw	
s8	32	0	rw	
s9	32	0	rw	
s10	32	0	rw	
s11	32	0	rw	
t3	32	0	rw	
t4	32	0	rw	
t5	32	0	rw	
t6	32	0	rw	
pc	32	0	rw	program counter

Table 13.1: Registers at level 1, type:Hart group:Core

13.1.2 User_Control_and_Status

Registers at level:1, type:Hart group:User_Control_and_Status

Name	Bits	Initial-Hex	RW	Description
cycle	32	0	r-	Cycle Counter
time	32	0	r-	Timer
instret	32	0	r-	Instructions Retired
hpmcounter3	32	0	r-	Performance Monitor Counter 3
hpmcounter4	32	0	r-	Performance Monitor Counter 4
hpmcounter5	32	0	r-	Performance Monitor Counter 5
hpmcounter6	32	0	r-	Performance Monitor Counter 6
hpmcounter7	32	0	r-	Performance Monitor Counter 7
hpmcounter8	32	0	r-	Performance Monitor Counter 8
hpmcounter9	32	0	r-	Performance Monitor Counter 9
hpmcounter10	32	0	r-	Performance Monitor Counter 10
hpmcounter11	32	0	r-	Performance Monitor Counter 11
hpmcounter12	32	0	r-	Performance Monitor Counter 12
hpmcounter13	32	0	r-	Performance Monitor Counter 13
hpmcounter14	32	0	r-	Performance Monitor Counter 14
hpmcounter15	32	0	r-	Performance Monitor Counter 15
hpmcounter16	32	0	r-	Performance Monitor Counter 16
hpmcounter17	32	0	r-	Performance Monitor Counter 17
hpmcounter18	32	0	r-	Performance Monitor Counter 18
hpmcounter19	32	0	r-	Performance Monitor Counter 19
hpmcounter20	32	0	r-	Performance Monitor Counter 20
hpmcounter21	32	0	r-	Performance Monitor Counter 21
hpmcounter22	32	0	r-	Performance Monitor Counter 22
hpmcounter23	32	0	r-	Performance Monitor Counter 23
hpmcounter24	32	0	r-	Performance Monitor Counter 24
hpmcounter25	32	0	r-	Performance Monitor Counter 25
hpmcounter26	32	0	r-	Performance Monitor Counter 26
hpmcounter27	32	0	r-	Performance Monitor Counter 27
hpmcounter28	32	0	r-	Performance Monitor Counter 28
hpmcounter29	32	0	r-	Performance Monitor Counter 29
hpmcounter30	32	0	r-	Performance Monitor Counter 30
hpmcounter31	32	0	r-	Performance Monitor Counter 31
cycleh	32	0	r-	Cycle Counter High
timeh	32	0	r-	Timer High
instreth	32	0	r-	Instructions Retired High
hpmcounter3h	32	0	r-	Performance Monitor High 3
hpmcounter4h	32	0	r-	Performance Monitor High 4
hpmcounter5h	32	0	r-	Performance Monitor High 5
hpmcounter6h	32	0	r-	Performance Monitor High 6
hpmcounter7h	32	0	r-	Performance Monitor High 7
hpmcounter8h	32	0	r-	Performance Monitor High 8
hpmcounter9h	32	0	r-	Performance Monitor High 9
hpmcounter10h	32	0	r-	Performance Monitor High 10
hpmcounter11h	32	0	r-	Performance Monitor High 11
hpmcounter12h	32	0	r-	Performance Monitor High 12
hpmcounter13h	32	0	r-	Performance Monitor High 13
hpmcounter14h	32	0	r-	Performance Monitor High 14

hpmcounter15h	32	0	r-	Performance Monitor High 15
hpmcounter16h	32	0	r-	Performance Monitor High 16
hpmcounter17h	32	0	r-	Performance Monitor High 17
hpmcounter18h	32	0	r-	Performance Monitor High 18
hpmcounter19h	32	0	r-	Performance Monitor High 19
hpmcounter20h	32	0	r-	Performance Monitor High 20
hpmcounter21h	32	0	r-	Performance Monitor High 21
hpmcounter22h	32	0	r-	Performance Monitor High 22
hpmcounter23h	32	0	r-	Performance Monitor High 23
hpmcounter24h	32	0	r-	Performance Monitor High 24
hpmcounter25h	32	0	r-	Performance Monitor High 25
hpmcounter26h	32	0	r-	Performance Monitor High 26
hpmcounter27h	32	0	r-	Performance Monitor High 27
hpmcounter28h	32	0	r-	Performance Monitor High 28
hpmcounter29h	32	0	r-	Performance Monitor High 29
hpmcounter30h	32	0	r-	Performance Monitor High 30
hpmcounter31h	32	0	r-	Performance Monitor High 31

Table 13.2: Registers at level 1, type:Hart group:User_Control_and_Status

13.1.3 Supervisor_Control_and_Status

Registers at level:1, type:Hart group:Supervisor_Control_and_Status

Name	Bits	Initial-Hex	RW	Description
sstatus	32	0	rw	Supervisor Status
sie	32	0	rw	Supervisor Interrupt Enable
stvec	32	0	rw	Supervisor Trap-Vector Base-Address
scounteren	32	0	rw	Supervisor Counter Enable
senvcfg	32	0	rw	Supervisor Environment Configuration
sscratch	32	0	rw	Supervisor Scratch
sepc	32	0	rw	Supervisor Exception Program Counter
scause	32	0	rw	Supervisor Cause
stval	32	0	rw	Supervisor Trap Value
sip	32	0	rw	Supervisor Interrupt Pending
satp	32	0	rw	Supervisor Address Translation and Protection
scontext	32	0	rw	Trigger Supervisor Context
mscontext	32	0	rw	Trigger Machine Context Alias

Table 13.3: Registers at level 1, type:Hart group:Supervisor_Control_and_Status

13.1.4 Machine_Control_and_Status

Registers at level:1, type:Hart group:Machine_Control_and_Status

Name	Bits	Initial-Hex	RW	Description
mstatus	32	0	rw	Machine Status
misa	32	40141105	rw	ISA and Extensions
medeleg	32	0	rw	Machine Exception Delegation
mideleg	32	0	rw	Machine Interrupt Delegation
mie	32	0	rw	Machine Interrupt Enable
mtvec	32	0	rw	Machine Trap-Vector Base-Address
mcounteren	32	0	rw	Machine Counter Enable
menvcfg	32	0	rw	Machine Environment Configuration
mstathsh	32	0	rw	Machine Status High

medeleg	32	0	rw	Machine Exception Delegation High
menvcfgh	32	0	rw	Machine Environment Configuration High
mcountinhibit	32	0	rw	Machine Counter Inhibit
mhpmevent3	32	0	rw	Machine Performance Monitor Event Select 3
mhpmevent4	32	0	rw	Machine Performance Monitor Event Select 4
mhpmevent5	32	0	rw	Machine Performance Monitor Event Select 5
mhpmevent6	32	0	rw	Machine Performance Monitor Event Select 6
mhpmevent7	32	0	rw	Machine Performance Monitor Event Select 7
mhpmevent8	32	0	rw	Machine Performance Monitor Event Select 8
mhpmevent9	32	0	rw	Machine Performance Monitor Event Select 9
mhpmevent10	32	0	rw	Machine Performance Monitor Event Select 10
mhpmevent11	32	0	rw	Machine Performance Monitor Event Select 11
mhpmevent12	32	0	rw	Machine Performance Monitor Event Select 12
mhpmevent13	32	0	rw	Machine Performance Monitor Event Select 13
mhpmevent14	32	0	rw	Machine Performance Monitor Event Select 14
mhpmevent15	32	0	rw	Machine Performance Monitor Event Select 15
mhpmevent16	32	0	rw	Machine Performance Monitor Event Select 16
mhpmevent17	32	0	rw	Machine Performance Monitor Event Select 17
mhpmevent18	32	0	rw	Machine Performance Monitor Event Select 18
mhpmevent19	32	0	rw	Machine Performance Monitor Event Select 19
mhpmevent20	32	0	rw	Machine Performance Monitor Event Select 20
mhpmevent21	32	0	rw	Machine Performance Monitor Event Select 21
mhpmevent22	32	0	rw	Machine Performance Monitor Event Select 22
mhpmevent23	32	0	rw	Machine Performance Monitor Event Select 23
mhpmevent24	32	0	rw	Machine Performance Monitor Event Select 24
mhpmevent25	32	0	rw	Machine Performance Monitor Event Select 25
mhpmevent26	32	0	rw	Machine Performance Monitor Event Select 26
mhpmevent27	32	0	rw	Machine Performance Monitor Event Select 27
mhpmevent28	32	0	rw	Machine Performance Monitor Event Select 28
mhpmevent29	32	0	rw	Machine Performance Monitor Event Select 29
mhpmevent30	32	0	rw	Machine Performance Monitor Event Select 30
mhpmevent31	32	0	rw	Machine Performance Monitor Event Select 31
mscratch	32	0	rw	Machine Scratch
mepc	32	0	rw	Machine Exception Program Counter
mcause	32	0	rw	Machine Cause
mtval	32	0	rw	Machine Trap Value
mip	32	0	rw	Machine Interrupt Pending
pmpcfg0	32	0	rw	Physical Memory Protection Configuration 0
pmpcfg1	32	0	rw	Physical Memory Protection Configuration 1
pmpcfg2	32	0	rw	Physical Memory Protection Configuration 2
pmpcfg3	32	0	rw	Physical Memory Protection Configuration 3
pmpcfg4	32	0	rw	Physical Memory Protection Configuration 4
pmpcfg5	32	0	rw	Physical Memory Protection Configuration 5
pmpcfg6	32	0	rw	Physical Memory Protection Configuration 6
pmpcfg7	32	0	rw	Physical Memory Protection Configuration 7
pmpcfg8	32	0	rw	Physical Memory Protection Configuration 8
pmpcfg9	32	0	rw	Physical Memory Protection Configuration 9
pmpcfg10	32	0	rw	Physical Memory Protection Configuration 10
pmpcfg11	32	0	rw	Physical Memory Protection Configuration 11
pmpcfg12	32	0	rw	Physical Memory Protection Configuration 12
pmpcfg13	32	0	rw	Physical Memory Protection Configuration 13
pmpcfg14	32	0	rw	Physical Memory Protection Configuration 14
pmpcfg15	32	0	rw	Physical Memory Protection Configuration 15
pmpaddr0	32	0	rw	Physical Memory Protection Address 0
pmpaddr1	32	0	rw	Physical Memory Protection Address 1
pmpaddr2	32	0	rw	Physical Memory Protection Address 2

pmpaddr3	32	0	rw	Physical Memory Protection Address 3
pmpaddr4	32	0	rw	Physical Memory Protection Address 4
pmpaddr5	32	0	rw	Physical Memory Protection Address 5
pmpaddr6	32	0	rw	Physical Memory Protection Address 6
pmpaddr7	32	0	rw	Physical Memory Protection Address 7
pmpaddr8	32	0	rw	Physical Memory Protection Address 8
pmpaddr9	32	0	rw	Physical Memory Protection Address 9
pmpaddr10	32	0	rw	Physical Memory Protection Address 10
pmpaddr11	32	0	rw	Physical Memory Protection Address 11
pmpaddr12	32	0	rw	Physical Memory Protection Address 12
pmpaddr13	32	0	rw	Physical Memory Protection Address 13
pmpaddr14	32	0	rw	Physical Memory Protection Address 14
pmpaddr15	32	0	rw	Physical Memory Protection Address 15
pmpaddr16	32	0	rw	Physical Memory Protection Address 16
pmpaddr17	32	0	rw	Physical Memory Protection Address 17
pmpaddr18	32	0	rw	Physical Memory Protection Address 18
pmpaddr19	32	0	rw	Physical Memory Protection Address 19
pmpaddr20	32	0	rw	Physical Memory Protection Address 20
pmpaddr21	32	0	rw	Physical Memory Protection Address 21
pmpaddr22	32	0	rw	Physical Memory Protection Address 22
pmpaddr23	32	0	rw	Physical Memory Protection Address 23
pmpaddr24	32	0	rw	Physical Memory Protection Address 24
pmpaddr25	32	0	rw	Physical Memory Protection Address 25
pmpaddr26	32	0	rw	Physical Memory Protection Address 26
pmpaddr27	32	0	rw	Physical Memory Protection Address 27
pmpaddr28	32	0	rw	Physical Memory Protection Address 28
pmpaddr29	32	0	rw	Physical Memory Protection Address 29
pmpaddr30	32	0	rw	Physical Memory Protection Address 30
pmpaddr31	32	0	rw	Physical Memory Protection Address 31
pmpaddr32	32	0	rw	Physical Memory Protection Address 32
pmpaddr33	32	0	rw	Physical Memory Protection Address 33
pmpaddr34	32	0	rw	Physical Memory Protection Address 34
pmpaddr35	32	0	rw	Physical Memory Protection Address 35
pmpaddr36	32	0	rw	Physical Memory Protection Address 36
pmpaddr37	32	0	rw	Physical Memory Protection Address 37
pmpaddr38	32	0	rw	Physical Memory Protection Address 38
pmpaddr39	32	0	rw	Physical Memory Protection Address 39
pmpaddr40	32	0	rw	Physical Memory Protection Address 40
pmpaddr41	32	0	rw	Physical Memory Protection Address 41
pmpaddr42	32	0	rw	Physical Memory Protection Address 42
pmpaddr43	32	0	rw	Physical Memory Protection Address 43
pmpaddr44	32	0	rw	Physical Memory Protection Address 44
pmpaddr45	32	0	rw	Physical Memory Protection Address 45
pmpaddr46	32	0	rw	Physical Memory Protection Address 46
pmpaddr47	32	0	rw	Physical Memory Protection Address 47
pmpaddr48	32	0	rw	Physical Memory Protection Address 48
pmpaddr49	32	0	rw	Physical Memory Protection Address 49
pmpaddr50	32	0	rw	Physical Memory Protection Address 50
pmpaddr51	32	0	rw	Physical Memory Protection Address 51
pmpaddr52	32	0	rw	Physical Memory Protection Address 52
pmpaddr53	32	0	rw	Physical Memory Protection Address 53
pmpaddr54	32	0	rw	Physical Memory Protection Address 54
pmpaddr55	32	0	rw	Physical Memory Protection Address 55
pmpaddr56	32	0	rw	Physical Memory Protection Address 56
pmpaddr57	32	0	rw	Physical Memory Protection Address 57
pmpaddr58	32	0	rw	Physical Memory Protection Address 58

pmpaddr59	32	0	rw	Physical Memory Protection Address 59
pmpaddr60	32	0	rw	Physical Memory Protection Address 60
pmpaddr61	32	0	rw	Physical Memory Protection Address 61
pmpaddr62	32	0	rw	Physical Memory Protection Address 62
pmpaddr63	32	0	rw	Physical Memory Protection Address 63
tselect	32	0	rw	Trigger Register Select
tdata1	32	f0000000	rw	Trigger Data 1
tdata2	32	0	rw	Trigger Data 2
tdata3	32	0	rw	Trigger Data 3
tinfo	32	100807c	rw	Trigger Info
tcontrol	32	0	rw	Trigger Control
mcontext	32	0	rw	Trigger Machine Context
mcycle	32	0	rw	Machine Cycle Counter
minstret	32	0	rw	Machine Instructions Retired
mhpcounter3	32	0	rw	Machine Performance Monitor Counter 3
mhpcounter4	32	0	rw	Machine Performance Monitor Counter 4
mhpcounter5	32	0	rw	Machine Performance Monitor Counter 5
mhpcounter6	32	0	rw	Machine Performance Monitor Counter 6
mhpcounter7	32	0	rw	Machine Performance Monitor Counter 7
mhpcounter8	32	0	rw	Machine Performance Monitor Counter 8
mhpcounter9	32	0	rw	Machine Performance Monitor Counter 9
mhpcounter10	32	0	rw	Machine Performance Monitor Counter 10
mhpcounter11	32	0	rw	Machine Performance Monitor Counter 11
mhpcounter12	32	0	rw	Machine Performance Monitor Counter 12
mhpcounter13	32	0	rw	Machine Performance Monitor Counter 13
mhpcounter14	32	0	rw	Machine Performance Monitor Counter 14
mhpcounter15	32	0	rw	Machine Performance Monitor Counter 15
mhpcounter16	32	0	rw	Machine Performance Monitor Counter 16
mhpcounter17	32	0	rw	Machine Performance Monitor Counter 17
mhpcounter18	32	0	rw	Machine Performance Monitor Counter 18
mhpcounter19	32	0	rw	Machine Performance Monitor Counter 19
mhpcounter20	32	0	rw	Machine Performance Monitor Counter 20
mhpcounter21	32	0	rw	Machine Performance Monitor Counter 21
mhpcounter22	32	0	rw	Machine Performance Monitor Counter 22
mhpcounter23	32	0	rw	Machine Performance Monitor Counter 23
mhpcounter24	32	0	rw	Machine Performance Monitor Counter 24
mhpcounter25	32	0	rw	Machine Performance Monitor Counter 25
mhpcounter26	32	0	rw	Machine Performance Monitor Counter 26
mhpcounter27	32	0	rw	Machine Performance Monitor Counter 27
mhpcounter28	32	0	rw	Machine Performance Monitor Counter 28
mhpcounter29	32	0	rw	Machine Performance Monitor Counter 29
mhpcounter30	32	0	rw	Machine Performance Monitor Counter 30
mhpcounter31	32	0	rw	Machine Performance Monitor Counter 31
mcycleh	32	0	rw	Machine Cycle Counter High
minstreth	32	0	rw	Machine Instructions Retired High
mhpcounter3h	32	0	rw	Machine Performance Monitor Counter High 3
mhpcounter4h	32	0	rw	Machine Performance Monitor Counter High 4
mhpcounter5h	32	0	rw	Machine Performance Monitor Counter High 5
mhpcounter6h	32	0	rw	Machine Performance Monitor Counter High 6
mhpcounter7h	32	0	rw	Machine Performance Monitor Counter High 7
mhpcounter8h	32	0	rw	Machine Performance Monitor Counter High 8
mhpcounter9h	32	0	rw	Machine Performance Monitor Counter High 9
mhpcounter10h	32	0	rw	Machine Performance Monitor Counter High 10
mhpcounter11h	32	0	rw	Machine Performance Monitor Counter High 11
mhpcounter12h	32	0	rw	Machine Performance Monitor Counter High 12
mhpcounter13h	32	0	rw	Machine Performance Monitor Counter High 13

mhpmpcounter14h	32	0	rw	Machine Performance Monitor Counter High 14
mhpmpcounter15h	32	0	rw	Machine Performance Monitor Counter High 15
mhpmpcounter16h	32	0	rw	Machine Performance Monitor Counter High 16
mhpmpcounter17h	32	0	rw	Machine Performance Monitor Counter High 17
mhpmpcounter18h	32	0	rw	Machine Performance Monitor Counter High 18
mhpmpcounter19h	32	0	rw	Machine Performance Monitor Counter High 19
mhpmpcounter20h	32	0	rw	Machine Performance Monitor Counter High 20
mhpmpcounter21h	32	0	rw	Machine Performance Monitor Counter High 21
mhpmpcounter22h	32	0	rw	Machine Performance Monitor Counter High 22
mhpmpcounter23h	32	0	rw	Machine Performance Monitor Counter High 23
mhpmpcounter24h	32	0	rw	Machine Performance Monitor Counter High 24
mhpmpcounter25h	32	0	rw	Machine Performance Monitor Counter High 25
mhpmpcounter26h	32	0	rw	Machine Performance Monitor Counter High 26
mhpmpcounter27h	32	0	rw	Machine Performance Monitor Counter High 27
mhpmpcounter28h	32	0	rw	Machine Performance Monitor Counter High 28
mhpmpcounter29h	32	0	rw	Machine Performance Monitor Counter High 29
mhpmpcounter30h	32	0	rw	Machine Performance Monitor Counter High 30
mhpmpcounter31h	32	0	rw	Machine Performance Monitor Counter High 31
mvendorid	32	0	r-	Vendor ID
marchid	32	0	r-	Architecture ID
mimpid	32	0	r-	Implementation ID
mhartid	32	0	r-	Hardware Thread ID
mconfigptr	32	0	r-	Configuration Data Structure

Table 13.4: Registers at level 1, type:Hart group:Machine_Control_and_Status

13.1.5 Integration support

Registers at level:1, type:Hart group:Integration_support

Name	Bits	Initial-Hex	RW	Description
LRSCAddress	32	ffffff	rw	LR/SC active lock address
LRSCSize	8	0	rw	LR/SC active access byte size
commercial	8	0	r-	Commercial feature in use
PTWStage	8	0	r-	PTW active stage from memory callback context (0:none 1:HS 2:VS 3:G)
PTWInputAddr	64	0	r-	PTW input address from memory callback context
PTWLevel	8	0	r-	PTW active level from memory callback context
ASYNCPE	8	0	r-	Asynchronous Event Pending & Enabled
HaltReason	32	0	r-	Bit field indicating halt reason
mask_pmpcfg0	64	ffffff ffffff	r-	Write mask for pmpcfg0
mask_pmpcfg1	32	ffffff	r-	Write mask for pmpcfg1
mask_pmpcfg2	64	ffffff ffffff	r-	Write mask for pmpcfg2
mask_pmpcfg3	32	ffffff	r-	Write mask for pmpcfg3
mask_pmpaddr0	64	ffffff ffffff	r-	Write mask for pmpaddr0
mask_pmpaddr1	64	ffffff ffffff	r-	Write mask for pmpaddr1
mask_pmpaddr2	64	ffffff ffffff	r-	Write mask for pmpaddr2
mask_pmpaddr3	64	ffffff ffffff	r-	Write mask for pmpaddr3
mask_pmpaddr4	64	ffffff ffffff	r-	Write mask for pmpaddr4
mask_pmpaddr5	64	ffffff ffffff	r-	Write mask for pmpaddr5
mask_pmpaddr6	64	ffffff ffffff	r-	Write mask for pmpaddr6
mask_pmpaddr7	64	ffffff ffffff	r-	Write mask for pmpaddr7
mask_pmpaddr8	64	ffffff ffffff	r-	Write mask for pmpaddr8
mask_pmpaddr9	64	ffffff ffffff	r-	Write mask for pmpaddr9
mask_pmpaddr10	64	ffffff ffffff	r-	Write mask for pmpaddr10

mask_pmpaddr11	64	ffffff ffffffff	r-	Write mask for pmpaddr11
mask_pmpaddr12	64	ffffff ffffffff	r-	Write mask for pmpaddr12
mask_pmpaddr13	64	ffffff ffffffff	r-	Write mask for pmpaddr13
mask_pmpaddr14	64	ffffff ffffffff	r-	Write mask for pmpaddr14
mask_pmpaddr15	64	ffffff ffffffff	r-	Write mask for pmpaddr15
PTWBankSelect	8	0	rw	select PTW bank (0 is stage 1, 1 is stage 2, 2-6 are stage 2 walks initiated by stage 1 level 0-4 entry lookups, respectively)
PTWBankValid	8	0	r-	bitmask of valid banks (0x01 is stage 1, 0x02 is stage 2, 0x04-0x40 are stage 2 walks initiated by stage 1 level 0-4 entry lookups, respectively)
PTWAddressValid	8	0	r-	bitmask of valid bits for each of PTWAddressL0...PTWAddressL4, PTWBase, PTWInput and PTWOutput in current bank
PTWValueValid	8	0	r-	bitmask of valid bits for each of PTWValueL0...PTWValueL4 in current bank
PTWAddressL0	64	0	r-	current bank PTW address, level 0
PTWAddressL1	64	0	r-	current bank PTW address, level 1
PTWAddressL2	64	0	r-	current bank PTW address, level 2
PTWAddressL3	64	0	r-	current bank PTW address, level 3
PTWAddressL4	64	0	r-	current bank PTW address, level 4
PTWValueL0	64	0	r-	current bank PTW value, level 0
PTWValueL1	64	0	r-	current bank PTW value, level 1
PTWValueL2	64	0	r-	current bank PTW value, level 2
PTWValueL3	64	0	r-	current bank PTW value, level 3
PTWValueL4	64	0	r-	current bank PTW value, level 4
PTWBase	64	0	r-	current bank PTW table base address
PTWInput	64	0	r-	current bank PTW input address
PTWOutput	64	0	r-	current bank PTW output address
PTWPgSize	64	0	r-	current bank PTW page size (Valid only when PTWOutput is valid)
PTWUD	64	0	-w	perform [H]U-mode page table walk for the given data address
PTWUX	64	0	-w	perform [H]U-mode page table walk for the given execute address
PTWSD	64	0	-w	perform [H]S-mode page table walk for the given data address
PTWSX	64	0	-w	perform [H]S-mode page table walk for the given execute address
pmp0cfg0	8	0	r-	
pmp1cfg0	8	0	r-	
pmp2cfg0	8	0	r-	
pmp3cfg0	8	0	r-	
pmp4cfg1	8	0	r-	
pmp5cfg1	8	0	r-	
pmp6cfg1	8	0	r-	
pmp7cfg1	8	0	r-	
pmp8cfg2	8	0	r-	
pmp9cfg2	8	0	r-	
pmp10cfg2	8	0	r-	
pmp11cfg2	8	0	r-	
pmp12cfg3	8	0	r-	
pmp13cfg3	8	0	r-	
pmp14cfg3	8	0	r-	
pmp15cfg3	8	0	r-	

Table 13.5: Registers at level 1, type:Hart group:Integration support